

- **Testing thermal irons.** The temperature of the iron must be tested before applying it to the hair. Use the following guidelines for testing a heated iron:
  - Clamp the iron closed on tissue paper or toweling and hold for 5 seconds.
  - If the paper scorches or turns brown, the iron is too hot.
  - Allow an overheated iron to cool before applying it to the hair.
  - Remember that fine, lightened, or badly damaged hair withstands less heat than normal hair.
- **Maintaining thermal irons.** Since the materials used to make irons can vary greatly, keep your irons clean and functioning properly by always following the manufacturer's directions.
- **Using thermal irons safely.** Consider the following precautions for using thermal irons:
  - Use thermal irons only after receiving instructions on their use.
  - Keep irons clean and do not overheat them.
  - Always test the iron temperature before using it on a client's hair.
  - Handle heated tools carefully and put them in a safe place to cool.
  - Do not place heated stoves near the station mirror as the heat can cause breakage.
  - Place a hard rubber comb between the client's scalp and the iron. Never use a metal comb.

**P 5-4** Manipulating Thermal Irons *pages 138–139*

## LEARN ABOUT ELECTROTHERAPY AND LIGHT THERAPY MACHINES

Electrotherapy and light therapy machines are important equipment items for performing skin care services in many men's spas and high-end, full service shops. Most manufacturers provide special classes to prepare you to use their equipment in a safe and effective manner. Although some basic applications are covered in this textbook, it is recommended that you acquire training in the use of specific machines before performing electrotherapy or light therapy services on clients.

### High-Frequency Machine

A **high-frequency machine** (HY-FREE-kwen-see muh-SHEEN) uses electricity to produce a high rate of oscillation within glass electrodes that are used in facial and scalp treatments. Discovered by Nikola Tesla, high-frequency applications warm the skin tissues to stimulate blood flow and to allow for better product absorption. Refer to Chapter 8 for more information.

**figure 5-58**  
Marcel iron.



**figure 5-59**  
Electric heater for pressing combs and conventional irons.



### Galvanic Machine

The galvanic machine is an apparatus with attachments designed to produce galvanic current. The main function of the machine is to introduce water-soluble products into the skin during a facial. Refer to Chapter 7 for more information.

### Other Accessories

Two additional accessories you may find in the barbershop are the comedone extractor and tweezers. A **comedone extractor** (KAHM-uh-dohn eks-TRAK-tur) is a metallic implement with a screw on attachment at each end. A fine needle point or lancet is situated at one end; the opposite end has a blunt prong with a hole in the center that is used to press out blackheads.

A *tweezer* is a metallic implement with two blunt prongs at one end that is used to pluck unsightly or ingrown hair. To remove a hair, position the tweezer close to the skin surface, grasp the hair, and pull it out in the direction from which it grows. For added comfort, the area can be pre-steamed to open the hair follicle for easier extraction.



#### STATE REGULATORY ALERT!

Be guided by your state barber board rules and regulations regarding the use of comedone extractors.

## Learn How to Remove Hair Clippings

After reading this section, you will be able to:

- LO 17 Identify ways to remove hair clippings.
- LO 18 Show how to perform two towel-wrapping methods.

Hair removal methods have changed over the years. The neck duster, once a traditional implement in barbershops, is no longer considered a safe and sanitary option for loose hair removal unless it is cleaned and disinfected after each use. Since a number of states prohibit the use of hair dusters, other methods are now used to remove loose hair and clippings. Some methods that are in compliance with state and local health codes are the following:

- A paper or cloth towel folded around the barber's hand
- Paper neck strips, although these may not facilitate a thorough dusting
- Vacuum systems, provided you clean and disinfect the nozzle attachment after each use and empty the container as hair accumulates within it

To complete this section, practice Procedure 5-3 to become proficient in two towel-wrapping methods used in the performance of barbering services.

# HONING THE RAZOR

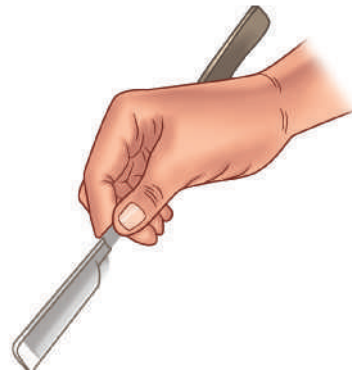
## MATERIALS, IMPLEMENTS, AND EQUIPMENT

- ☐ Conventional straight razor
- ☐ Shaving lather
- ☐ Hone
- ☐ Water

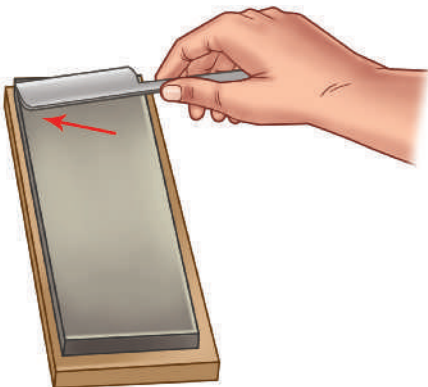
## PROCEDURE



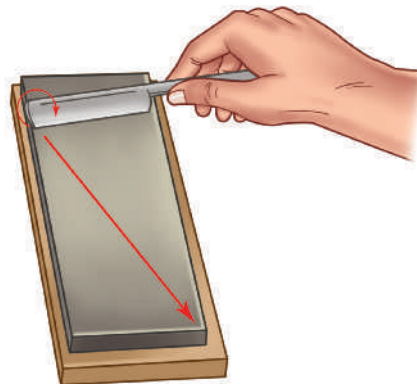
- 1 Position the hone by placing it firmly and flatly on a hard, smooth surface.



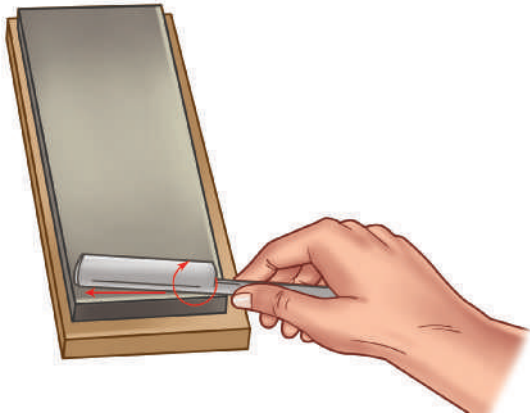
- 2 Grasp the razor handle comfortably in your dominant hand as follows:
- a. Rest the index finger on top of the side part of the shank.
  - b. Rest the ball of the thumb at the joint.
  - c. Place the second finger at the back of the razor near the edge of the shank.
  - d. Fold the remaining fingers around the handle to permit easy turning of the razor.



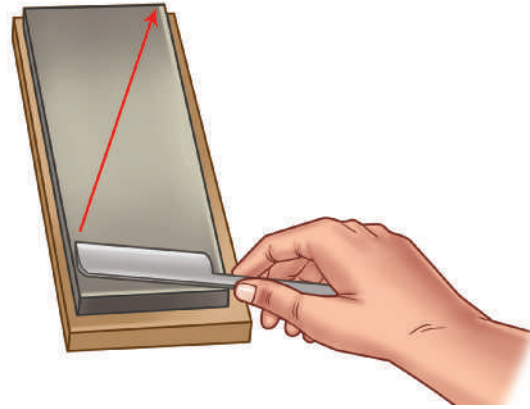
- 3 *First position and stroke:* Place the razor on its back on the upper far left corner of the hone.



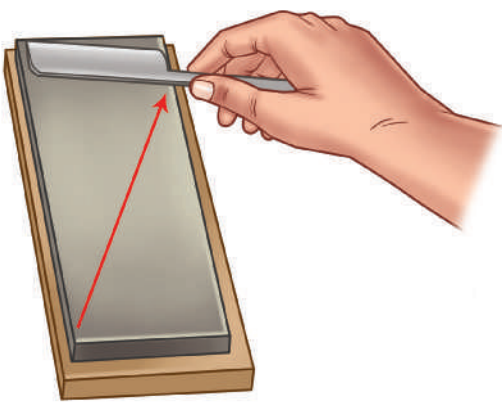
- 4 Roll the razor, using the thumb and index finger to position the blade edge flat against the hone and facing toward you. Draw the blade diagonally across the hone from heel to point.



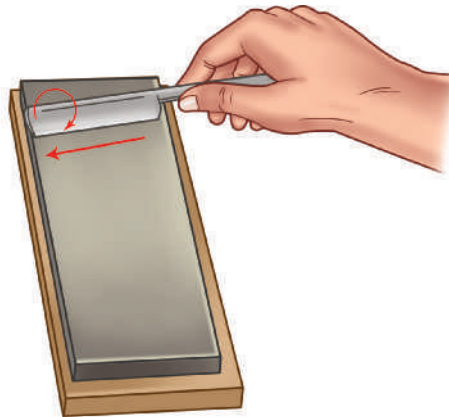
- 5 Turn the razor on its back and slide it toward the bottom left corner of the hone to position it for the second position and stroke.



- 6 *Second position and stroke:* Begin at lower left corner of the hone with the blade edge flat against the hone and facing away from you.



- 7 Stroke the blade diagonally toward the upper right corner of the hone.



- 8 Turn the razor on its back, slide it toward the upper left corner of the hone, and roll it into position for the next stroke.

- 9 Repeat the strokes in a slow and rhythmic manner with equal pressure. If the razor is very dull, use firm pressure during the first honing strokes and then decrease the pressure as the razor takes an edge.

- 10 Test the edge by lightly passing it over a thumbnail moistened with water or lather.

# STROPPING THE RAZOR

## MATERIALS, IMPLEMENTS, AND EQUIPMENT

- ☐ Barber chair
- ☐ Combination strop
- ☐ Conventional straight razor

## PROCEDURE

- 1 Clip the strop onto the arm of the barber chair. Hold the end of the strop firmly in the left hand on a slight diagonal from the chair and as high as is comfortable.
- 2 Grasp the razor firmly in the right hand with the thumb and index finger at the shank and the handle in your palm.



- 3 *First stroke:* Start the stroke at the top edge of the strop closest to the hand. Using a long, diagonal stroke with even pressure from the heel to the point, draw the razor perfectly flat, with back leading, straight over the surface. Bear down just heavily enough to feel the razor draw.



- 4 *Second stroke:* When the first stroke is completed, turn the razor on the back of the blade by rolling it between the fingers without turning the hand.



- 5 Draw the razor away from the chair toward you to complete the second stroke. Repeat strokes as necessary.
- 6 Make a final test of the razor prior to shaving on the moistened tip of the thumb.



# TOWEL WRAPPING

## MATERIALS, IMPLEMENTS, AND EQUIPMENT

- 100 percent cotton terry cloth or barber towel
- Disposable 12" × 24" professional paper towels

## PROCEDURE: CLOTH TOWEL WRAP



- 1 Grasp the towel lengthwise.



- 2 Holding your right hand in front of you, draw the upper edge of the towel across the palm of the hand; then grasp the towel end and twist.



- 3 Wrap the twisted ends of the towel around the back of the hand and bring over the inside of the wrist.

## PROCEDURE: PAPER TOWEL WRAP

- 4 Hold the ends of the towel while in use to prevent them from flapping in the client's face.



- 5 Grasp the towel lengthwise.



- 6 Fold down the top third of the towel toward you.



- 7 Holding the towel at one end, insert the two middle fingers into the fold; maintain your grip on the towel with the thumb, index finger, and fourth finger.



- 8 Bring the top edge around the back of the hand and secure with the fourth finger.



- 9 Grasp the towel end and shift to a diagonal position.



- 10 Wrap the remaining towel length around the back of the hand and insert thumb into the fold.



- 11 Continue wrapping motion around the thumb.



- 12 Tuck the towel end into wrap at the back of the hand.

# MANIPULATING THERMAL IRONS

## MATERIALS, IMPLEMENTS, AND EQUIPMENT

- ☐ Comb
- ☐ Electric curling iron
- ☐ Mannequin head and stand

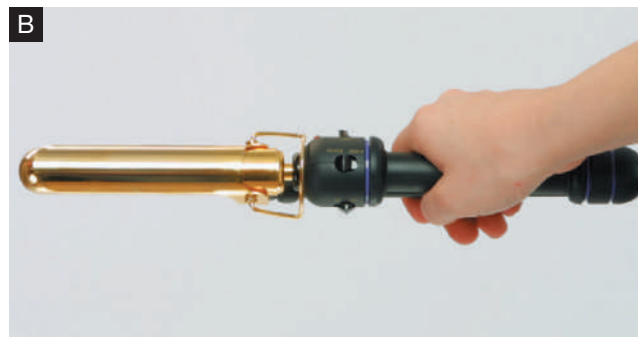
## PROCEDURE

- 1 Part off and comb a section of hair for curling.



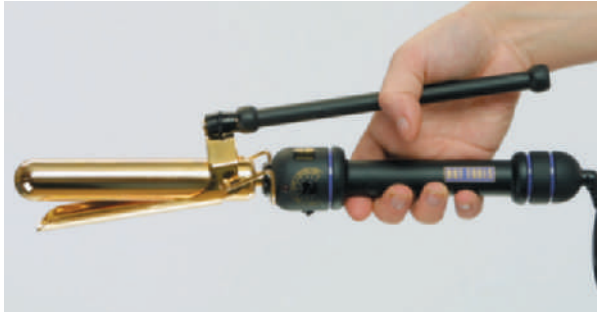
- 2 Grasp the handle with your dominant hand and open the clamp with your little finger.

- 3 Guide the iron so the hair section lies in between the clamp and the barrel of the iron.



- 4 Shift your thumb to the top of the iron, close the clamp, and make a one-quarter turn downward.





5 Use your thumb to open the clamp slightly to continue rotating the iron to a half turn and close the clamp.



6 Rotate the iron as you continue to open and close the clamp until you have completed a full turn of the iron.



7 Guide the remaining hair section into the center of the curl by opening and closing the clamp while rotating the iron.



8 Use a comb to secure the section while opening the clamp slightly to slide the iron out of the curl.

## REVIEW QUESTIONS

- 1 List the principal *tools of the trade* used in barbering.
- 2 What style of comb is generally used for haircutting?
- 3 Taper combs are especially effective for performing which three services?
- 4 Identify the parts of haircutting shears.
- 5 What is the difference between German and French shears?
- 6 How are the lengths of shears usually measured? Which sizes are used most often?
- 7 Hollow-ground blades are features of what type of shear?
- 8 What are thinning and blending shears used for?
- 9 List three types of clipper motors.
- 10 List the visible parts of an adjustable-blade electric clipper.
- 11 Identify the detachable clipper blade size that produces the closest cut.
- 12 Why should you never set outliner blades flush with each other?
- 13 Name two types of straight razors.
- 14 List the basic parts of a straight razor.
- 15 Explain the purpose of a hone.
- 16 List three types of hones.
- 17 Explain the first and second positions and strokes used to hone a razor.
- 18 Explain the purpose of a strop.
- 19 What type of strop is considered the best for stropping a conventional straight razor?
- 20 Explain the first and second strokes used to strop a razor.
- 21 List three methods of removing loose hair from a client's face and neck other than a neck brush.
- 22 What does the word *thermal* mean?
- 23 Explain the main function of the galvanic machine.
- 24 Identify two services that high-frequency machines might be used for.
- 25 What is the function of a comedone extractor?

## CHAPTER GLOSSARY

|                                                                            |        |                                                                                                                                                               |
|----------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>blades</b> (BLAYDZ)                                                     | p. 117 | the cutting parts of the clippers, usually manufactured from high-quality carbon steel and available in a variety of styles and sizes                         |
| <b>canvas strop</b> (can-viss strohp)                                      | p. 125 | usually one side of a combination strop made of linen or silk, woven into a fine or coarse texture that removes metal burrs or imbrications left after honing |
| <b>cast shears</b>                                                         | p. 108 | shears that have been made by heating steel to a liquid form for pouring into a mold                                                                          |
| <b>changeable-blade straight razor</b> (CHAYNJE-able BLAYD STRAYT RAY-zor) | p. 119 | a type of straight razor that uses changeable, disposable blades                                                                                              |
| <b>clippers</b> (KLIP-uhrz)                                                | p. 114 | electric haircutting tools with a single adjustable-blade or detachable-blade system                                                                          |
| <b>comedone extractor</b> (KAHM-uh-dohn eks-TRAK-tur)                      | p. 132 | an implement used to extract blackheads                                                                                                                       |

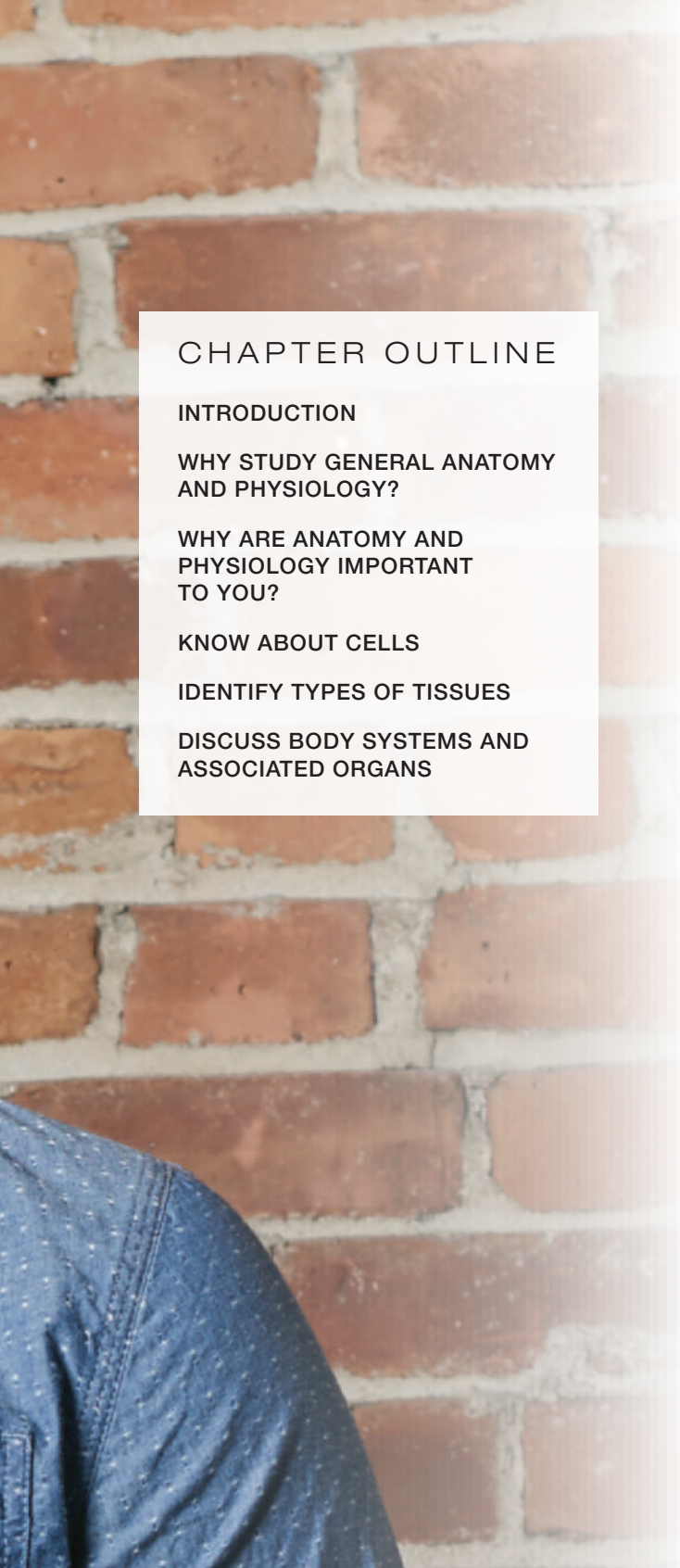
|                                                                        |        |                                                                                                                                                                                                    |
|------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>conventional straight razor</b><br>(kun-VEN-shun-ul STRAYT RAY-zor) | p. 119 | a razor made of a hardened steel blade that requires honing and stropping to produce a cutting edge                                                                                                |
| <b>forged shears</b> (FORJed SHEERZ)                                   | p. 109 | shears that have been made by working heated metal into a finished shape through the processes of hammering or compression                                                                         |
| <b>guards</b>                                                          | p. 117 | plastic or hard rubber comb attachments that fit over clipper blades to minimize the amount of hair being cut with the clippers; or a metal shield applied over a haircutting razor for protection |
| <b>high-frequency machine</b><br>(HY-FREE-kwen-see<br>muh-SHEEN)       | p. 131 | a machine that produces a high rate of oscillation or Tesla current for the purpose of stimulating scalp, facial, and body tissues                                                                 |
| <b>hone</b> (HOHN)                                                     | p. 123 | a sharpening block manufactured from rock or synthetic materials and used to create a cutting edge on conventional straight razors                                                                 |
| <b>palming the comb</b> (PAHM-ing<br>THE KOM)                          | p. 113 | the technique used to hold the comb in the hand opposite of the hand that is cutting with the shears                                                                                               |
| <b>palming the shears</b><br>(PAHM-ing THE SHEERZ)                     | p. 113 | the technique used to hold shears in a safe manner while combing through or otherwise working with hair                                                                                            |
| <b>Russian strop</b> (RUSH-an strohp)                                  | p. 125 | a type of cowhide strop that is considered to be one of the best and that requires breaking in                                                                                                     |
| <b>set of the shears</b>                                               | p. 110 | the manner in which the blades and shanks of the shears align with each other and are joined at the tension screw or rivet                                                                         |
| <b>shell strop</b>                                                     | p. 125 | a type of horsehide strop made from the muscular rump area of the horse and considered to be the best strop for use by barbers                                                                     |
| <b>strop</b>                                                           | p. 123 | an elongated piece of leather or other materials used to finish the edge of conventional straight razors to a smooth, whetted cutting edge                                                         |
| <b>taper comb</b> (TAY-pur KOM)                                        | p. 106 | used for cutting or trimming hair when a gradual blending from short to longer is required within the haircut                                                                                      |
| <b>texturizing shears</b><br>(TEKS-churyz-ing SHEERZ)                  | p. 111 | shears with 14 to 28 medium width teeth that produce visible differences in hair lengths to add texture and volume                                                                                 |
| <b>thermal styling tools</b><br>(THUR-mul STYL-ing TOOLS)              | p. 130 | refers to tools that produce heat for hairstyling purposes                                                                                                                                         |
| <b>thinning shears</b> (THIN-ing<br>SHEERZ)                            | p. 111 | also known as blending shears; shears with 30 to 50 thin, narrowly spaced teeth that eliminate visible lines in the hair, used for blending hair ends and removing bulk or weight                  |
| <b>trimmers</b> (TRIM-uhrz)                                            | p. 114 | small clippers, also known as outliners and edgers, used for detail, precision design, and fine finish work after a haircut or beard trim                                                          |



6



# GENERAL ANATOMY AND PHYSIOLOGY



## CHAPTER OUTLINE

### INTRODUCTION

WHY STUDY GENERAL ANATOMY  
AND PHYSIOLOGY?

WHY ARE ANATOMY AND  
PHYSIOLOGY IMPORTANT  
TO YOU?

KNOW ABOUT CELLS

IDENTIFY TYPES OF TISSUES

DISCUSS BODY SYSTEMS AND  
ASSOCIATED ORGANS

# LEARNING OBJECTIVES

After completing this chapter, you will be able to:

## LO<sup>1</sup>

Define and explain the importance of anatomy, physiology, and histology to the barbering profession.

## LO<sup>2</sup>

Describe cells, their structure, and their reproduction.

## LO<sup>3</sup>

Identify and define the types of tissues found in the body.

## LO<sup>4</sup>

Define organs and body systems.

## LO<sup>5</sup>

Name the main body systems and explain their basic functions.



# Introduction

Barbers are licensed to touch and perform services on clients in ways that are not permitted in many other occupations. This is an important responsibility, and as a barber, you should consider it an honor to be able to aid others in achieving a greater sense of well-being. To earn this responsibility, you need to have a solid understanding of the anatomy and physiology of the human body.

why study

## GENERAL ANATOMY AND PHYSIOLOGY?

**Barbers should study and have a thorough understanding of general anatomy and physiology because:**

- Understanding how the human body functions as an integrated whole is a key component in understanding how a client's hair and skin may react to various treatments and services.
- Being able to recognize the difference between what is considered normal and what is considered abnormal in the body will assist you in knowing what requires referral to a physician.
- Understanding the bone, muscle, and nerve structure of the human body will help you determine how to provide services that are appropriate for your client.

## Why Are Anatomy and Physiology Important to You?

After reading this section, you will be able to:

- LO 1** Define and explain the importance of anatomy, physiology, and histology to the barbering profession.

This high-level overview of human anatomy and physiology will provide you with a basic knowledge of the structure and functions of the human body to assist your performance in the barbering field. This chapter focuses on the anatomy and physiology that is pertinent to the practice of barbering, such as the anatomical structure of the head that provides a foundation for designing haircut and beard styles or performing a proper shave. This chapter will provide you with the definitions and “map” of the human body as a point of reference to be used when you discuss specific services later in the text.

**Anatomy** (ah-NAT-ah-mee) is the study of the human body structures that can be seen with the naked eye and how the body parts are organized; it is the science of the structure of organisms or of their parts.

**Physiology** (fiz-ih-OL-oh-gee) is the study of the functions and activities performed by the body's structures. The ending *-ology* means “study of.”

Barbers should be aware of the ways in which massage manipulations, heat, or absorptive products used in the practice of barbering might affect physiological activities of the body, such as increasing the circulation. Barbers should also be aware of certain physiological stresses caused by prolonged standing or repetitive movements that may affect their own skeletal or circulatory systems, as well as the preventative measures that can help minimize the occurrence of associated occupational disorders (see Chapter 3).

**DID YOU KNOW?**  
**Histology** (his-TAHL-uh-jee), also known as *microscopic anatomy* (mi-kroh-SKAHP-ik ah-NAT-ah-mee), is the study of tiny structures found in living tissues.

## Know about Cells

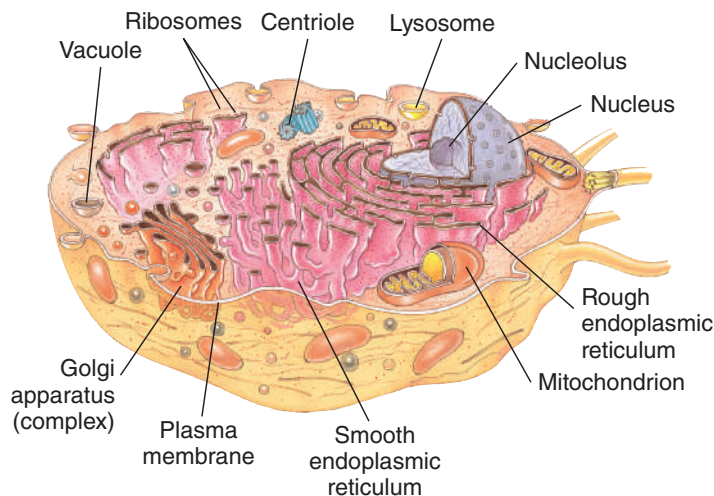
After reading this section, you will be able to:

**LO 2** Describe cells, their structure, and their reproduction.

**Cells** (SELLZ) are the basic units of all living things, from bacteria to plants to animals, including human beings. Without cells, life does not exist. As a basic functional unit, the cell is responsible for carrying on all life processes. There are trillions of cells in the human body, and they vary widely in size, shape, and purpose.

### BASIC STRUCTURE OF THE CELL

The cells of all living things are composed of a substance called **protoplasm** (PROH-toh-plaz-um), a colorless jellylike substance found inside cells in which food elements such as proteins, fats, carbohydrates, mineral salts, and water are present. You can visualize the protoplasm of a cell as being similar to raw egg white. In addition to protoplasm, most cells also include a nucleus, cytoplasm, and the cell membrane (**Figure 6-1**).



**figure 6-1**  
Basic structure of the cell.

The **nucleus** (NOO-klee-us) is the dense, active protoplasm found in the center of the cell; it plays an important part in cell reproduction and metabolism. You can visualize the nucleus as the yolk in the middle of a raw egg.

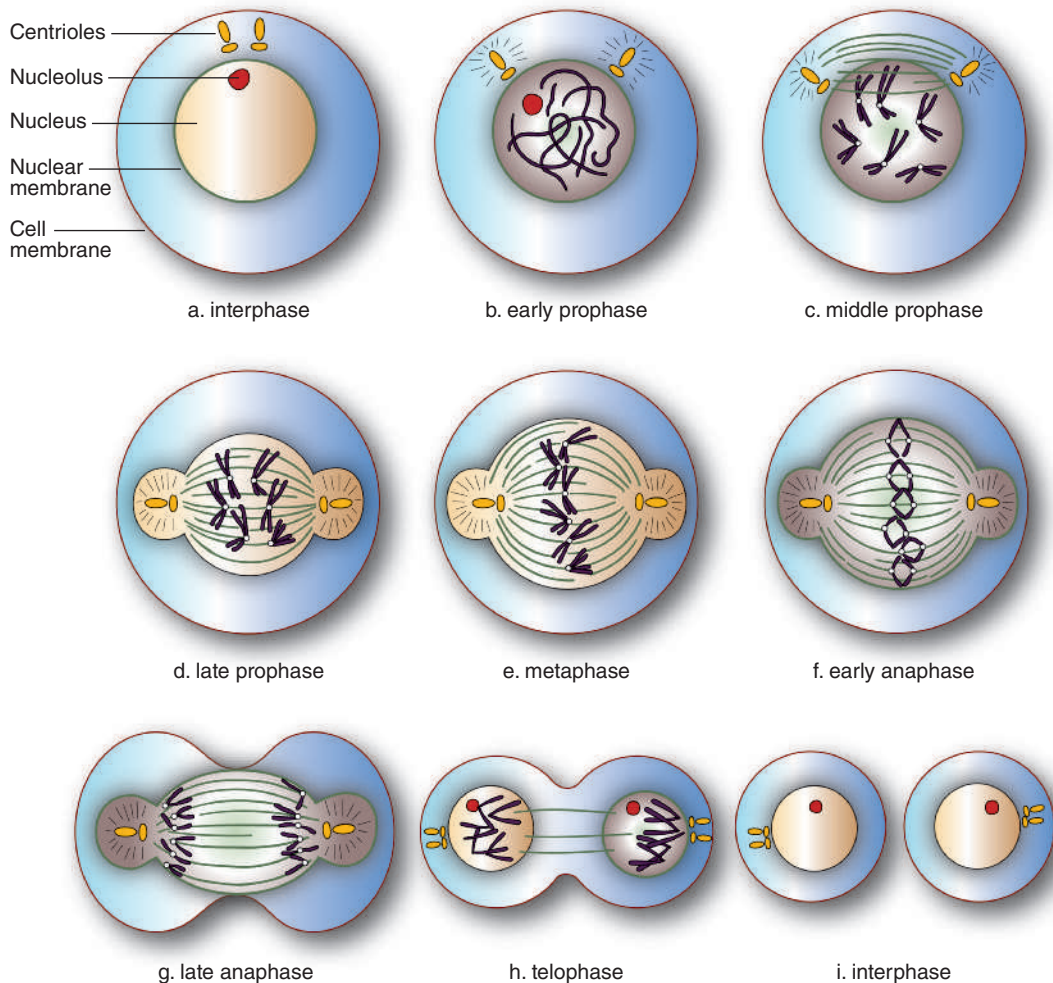
The **cytoplasm** (sy-toh-PLAZ-um) is the watery fluid that surrounds the nucleus of the cell and is needed for growth, reproduction, and self-repair. It is part of the protoplasm of the cell.

The **cell membrane** (SELL mem-brain) is the cell wall that encloses the protoplasm and permits soluble substances to enter and leave the cell.

## CELL REPRODUCTION AND DIVISION

Cells have the ability to reproduce, thus providing new cells for the growth and replacement of worn or injured ones. **Mitosis** (my-TOH-sis) is the usual process of cell reproduction of human tissues that occurs when the cell divides into two identical cells called *daughter cells*. Two small structures near the nucleus called **centrioles** (SEN-tree-olz) move to each side during the mitosis process to help divide the cell. As long as conditions are favorable, the cell will grow and reproduce. Favorable conditions include an adequate supply of food, oxygen, and water; suitable temperatures; and the ability to eliminate waste products. If conditions become unfavorable, the cell will become impaired or may die. Unfavorable conditions include toxins (poisons), disease, and injury (**Figure 6-2**).

**figure 6-2**  
Phases of mitosis.



# Identify Types of Tissues

After reading this section, you will be able to:

**LO3** Identify and define the types of tissues found in the body.

**Tissue** (TISH-oo) is a collection of similar cells that perform a particular function. Each kind of tissue has a specific function and can be recognized by its characteristic appearance. Body tissues are composed of large amounts of water, along with various other substances. There are four types of tissue in the body:

- **Connective tissue** (kuh-NEK-tiv TISH-oo) is fibrous tissue that binds together, protects, and supports the various parts of the body. Examples of connective tissue are bone, cartilage, ligaments, tendons, blood, lymph, and **adipose tissue** (ADD-ih-pohz TISH-oo), a technical term for fat. Adipose tissue gives smoothness and contour to the body.
- **Epithelial tissue** (ep-ih-THÉE-lee-ul TISH-oo) is a protective covering on body surfaces, such as skin, mucous membranes, the tissue inside the mouth, the lining of the heart, digestive and respiratory organs, and the glands.
- **Muscle tissue** contracts and moves various parts of the body.
- **Nerve tissue** carries messages to and from the brain and controls and coordinates all bodily functions. Nerve tissue is composed of special cells known as *neurons* that make up the nerves, brain, and spinal cord.



## DID YOU KNOW?

The skin is the body's largest organ.

# Discuss Body Systems and Associated Organs

After reading this section, you will be able to:

**LO4** Define organs and body systems.

**LO5** Name the main body systems and explain their basic functions.

**Organs** are structures composed of specialized tissues designed to perform specific functions in plants and animals. During the development of a fetus, tissues are “assigned” to specific functions in the body and they develop specifically for those functions. For example, lung tissue would not work as a part of the brain as it is designed to serve a specific function in the lungs. **Body systems** (BAHD-ee SYS-tums), also known as *systems*, are groups of body organs acting together to perform one or more functions (Figure 6-3). Table 6-1 outlines the body systems, indicating the functions of the system and the major organs that are associated with that system.

figure 6-3

Life is built from the carbon atom.

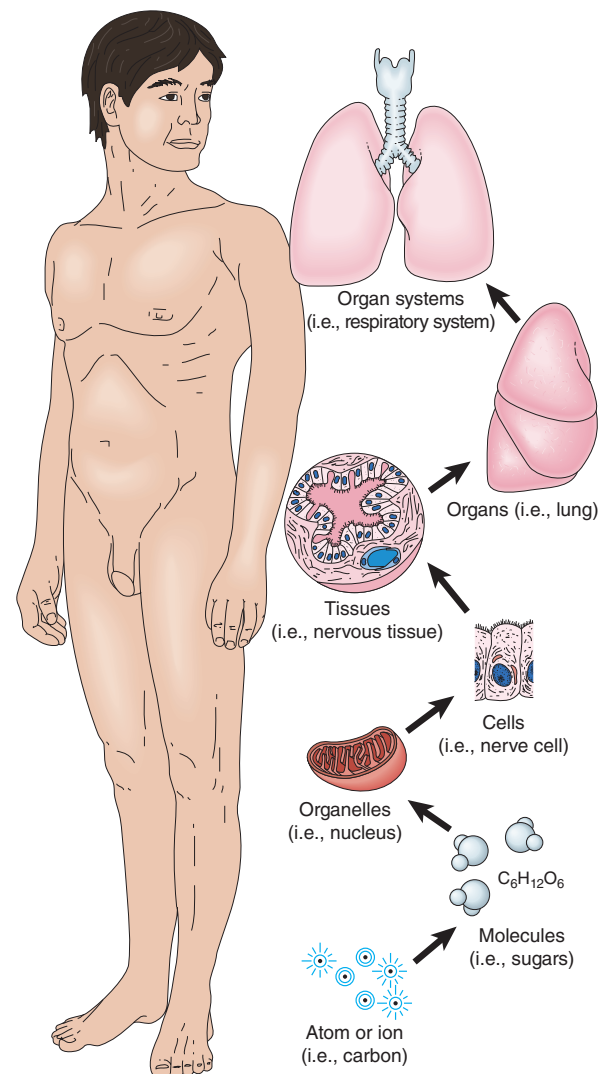




table 6-1

THE ELEVEN BODY SYSTEMS, THEIR FUNCTIONS, AND MAJOR ORGANS

| Body Systems                                           | Function                                                                                                    | Major Organs                                                     |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Circulatory                                            | Controls movement of blood throughout the body                                                              | Heart, blood vessels                                             |
| <b>Digestive</b><br>(dy-JES-tiv)<br>(gastrointestinal) | Breaks down food into nutrients or waste for nutrition or excretion                                         | Stomach, intestines, salivary and gastric glands                 |
| Endocrine                                              | Controls hormone levels that determine growth, development, sexual function, and health of entire body      | Endocrine glands, hormones                                       |
| <b>Excretory</b><br>(EKS-kruh-toh-ree)                 | Eliminates waste from the body reducing buildup of toxins                                                   | Kidneys, liver, skin, large intestines, lungs                    |
| Integumentary                                          | Provides protective covering and regulates body temperature                                                 | Skin, oil and sweat glands, hair, nails                          |
| Immune (lymphatic)                                     | Protects the body from disease by developing immunity and destroying pathogens and toxins                   | Lymph, lymph nodes, thymus gland, spleen                         |
| Muscular                                               | Covers, shapes, and holds the skeleton in place. Muscles contract to allow for movement of body structures. | Muscles, connective tissues                                      |
| Nervous                                                | Coordinates with all other body systems, allowing them to work efficiently and react to the environment     | Brain, spinal cord, nerves, eyes                                 |
| Reproductive                                           | Produces offspring and allows for transfer of genetic material. Differentiates between the sexes            | Female: ovaries, uterus, vagina<br>Male: testes, prostate, penis |
| <b>Respiratory</b> (RES-puh-rah-tor-ee)                | Makes blood and oxygen available to body structures through respiration; eliminates carbon dioxide          | Lungs, air passages                                              |
| Skeletal                                               | Forms the physical foundation of the body: 206 bones that are connected by movable and immovable joints     | Bones, joints                                                    |



**DID YOU KNOW?**

You have over 230 movable and semi-movable joints in your body.

REVIEW THE SKELETAL SYSTEM

The **skeletal system** (SKEL-uh-tul SIS-tum) forms the physical foundation of the body and is composed of 206 bones that vary in size and shape and are connected by movable and immovable joints. **Osteology** (ahs-tee-AHL-oh-jee) is the study of the anatomy, structure, and function of the bones. **Os** (AHS) means “bone.” It is used as a prefix in many medical terms, such as osteoarthritis, a joint disease.

Except for the tissue that forms the major part of the teeth, bone is the hardest tissue in the body. It is composed of connective tissue consisting of about one-third organic matter, such as cells and blood, and two-thirds minerals, mainly calcium carbonate and calcium phosphate.

The primary functions of the skeletal system are to:

- Give shape and support to the body.
- Protect various internal structures and organs.
- Serve as attachments for muscles and act as levers to produce body movement.

- Help produce both white and red blood cells (one of the functions of bone marrow).
- Store most of the body's calcium supply, as well as phosphorus, magnesium, and sodium.

A **joint** is the connection between two or more bones of the skeleton. There are two types of joints: movable, such as elbows, knees, and hips, and immovable, such as the joints found in the pelvis and skull, which allow little or no movement. There are exceptions to this condition such as child-birth, where special hormones allow for flexibility of the pelvic joints.

### **Bones of the Skull**

The skull is the skeleton of the head and is divided into two parts:

- **Cranium** (KRAY-nee-um). An oval, bony case that protects the brain.
- **Facial skeleton** (FAY-shul skell-uh-tin). The framework of the face that is composed of 14 bones.

### **Bones of the Cranium**

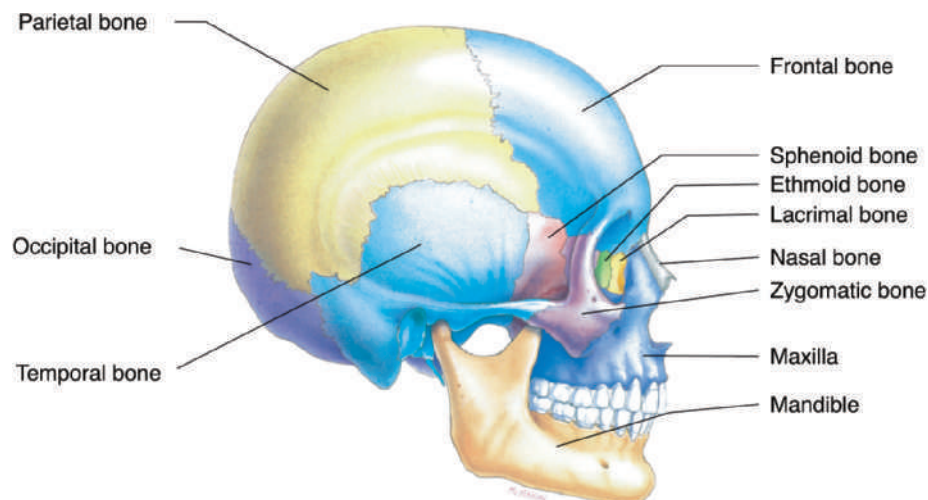
The following are the eight bones of the cranium (**Figure 6-4**):

- **Occipital bone** (ahk-SIP-it-ul BOHN). Hindmost bone of the skull, below the parietal bones; forms the back of the skull above the nape.
- **Parietal bones** (puh-RY-ate-ul BOHNZ). Bones that form the sides and top of the cranium. There are two parietal bones.
- **Frontal bone** (FRUNT-ul BOHN). Bone that forms the forehead.
- **Temporal bones** (TEM-puh-rul BOHNZ). Bones that form the sides of the head in the ear region. There are two temporal bones.
- **Ethmoid bone** (ETH-moyd BOHN). Light spongy bone between the eye sockets; forms part of the nasal cavities.
- **Sphenoid bone** (SFEE-noyd BOHN). Bone that joins all of the bones of the cranium together.

The ethmoid and sphenoid bones are not affected when performing services or giving a massage.

**DID YOU KNOW?**  
People often complain of joint pain; however, the pain is usually caused by inflammation of the tissue surrounding the joint and not by the joint itself.

**figure 6-4**  
Bones of the cranium and the face.



## Bones of the Face

There are 14 bones of the face, but those listed below are the most involved in the practice of barbering:

- **Nasal bones** (NAY-zul BONZ). Bones that form the bridge of the nose. There are two nasal bones.
- **Lacrimal bones** (LAK-ruh-mul BONZ). Small, thin bones located at the front inner wall of the orbits (eye sockets). There are two lacrimal bones.
- **Zygomatic bones** (zy-goh-MAT-ik BONZ), also known as *malar bones* or *cheekbones*. Bones that form the prominence of the cheeks. There are two zygomatic bones.
- **Maxillae** (mak-SIL-ee) (singular: maxilla [mak-SIL-uh]). Bones of the upper jaw. There are two maxillae.
- **Mandible** (MAN-duh-bul). Lower jawbone; largest and strongest bone of the face.



### DID YOU KNOW?

Painful inflammation involving the carpus (wrist) area can be caused by repetitive motions, such as flexing your wrist excessively or locking it in a bent position. Keeping the wrist straight can help prevent these injuries.

## Bones of the Neck

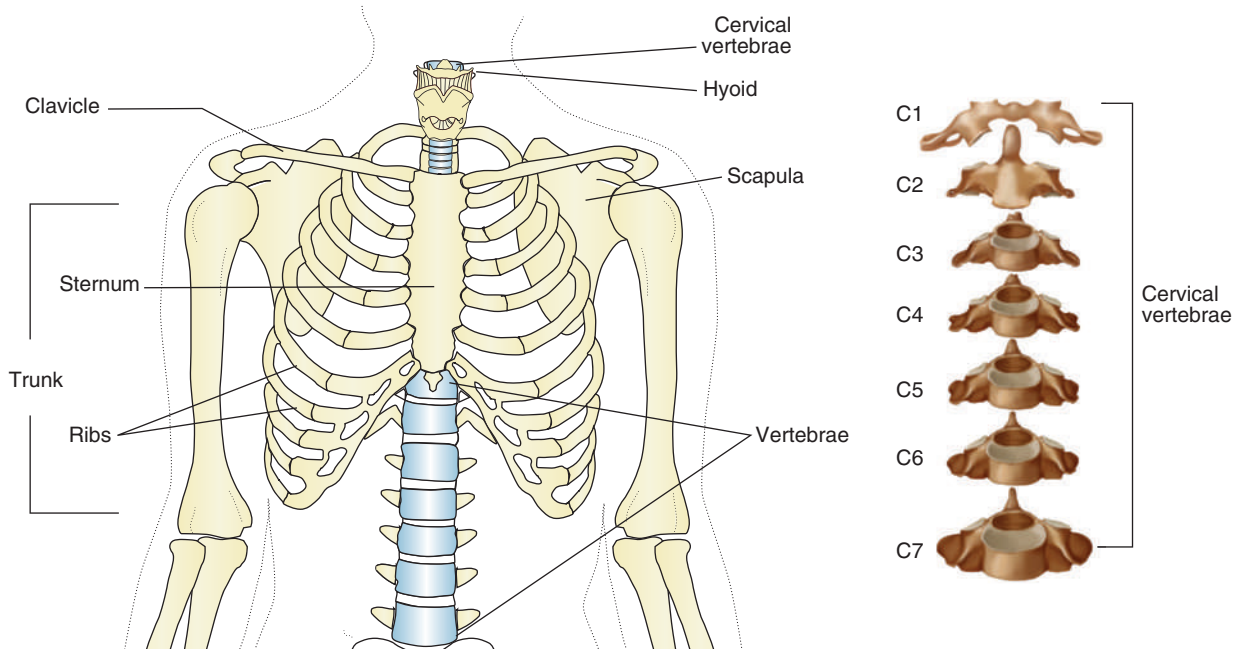
The main bones of the neck are the following:

- **Hyoid bone** (HY-oyd BOHN). U-shaped bone at the base of the tongue that supports the tongue and its muscles.
- **Cervical vertebrae** (SUR-vih-kul VURT-uh-bray). The seven bones of the top part of the vertebral column, located in the neck region ([Figure 6-5](#)).

## Bones of the Chest, Shoulder, and Back

The **thorax** (THOR-aks), or chest, is an elastic, bony cage consisting of the sternum, clavicle, scapula, and 12 pairs of ribs. This framework serves as a protective covering for the heart, lungs, and other delicate internal organs.

**figure 6-5**  
Bones of the neck, shoulders, and back.



## Bones of the Shoulder, Arm, and Hand

Some of the major causes of chronic pain for barbers are related to the repetitive action of the shoulders, arms, and hands. The important bones of the shoulders, arms, and hands that barbers should know about include the following (Figures 6-6 and 6-7):

- The **humerus** (HYOO-muh-rus) is the largest bone of the arm and extends from the shoulder to the elbow.
- The **ulna** (UL-nuh) is the inner and larger bone of the forearm, forms the elbow, and is located along the same side of the arm as the little finger.
- The **radius** (RAY-dee-us) is the smaller bone on the thumb side of the forearm.
- The **carpus** (KAR-pus) or wrist is a flexible joint composed of eight small, irregular bones held together by ligaments.
- The **metacarpals** (met-uh-KAR-pulz) are the bones of the palm, consisting of five slender bones between the carpus and the phalanges.
- The **phalanges** (FA-lanj-eez) are the bones of the fingers, consisting of three in each finger and two in the thumb, totaling 14 bones in each hand.

## REVIEW THE MUSCULAR SYSTEM

The **muscular system** (MUS-kyuh-lur SIS-tum) covers, shapes, and supports the skeleton, and its function is to help produce movement within the body.

**Myology** (my-AHL-uh-jee) is the study of the structure, functioning, and diseases of the muscles.

The muscular system consists of over 600 large and small muscles that comprise 40 to 50 percent of the body's weight. Muscles are composed of contractile, fibrous tissues that stretch and contract to facilitate the action of various body movements. The muscular system relies upon the skeletal and nervous systems for its activities and proper operation.

There are three types of muscular tissues:

1. Striated (striped) or voluntary muscles contain nerves, are attached to bones, and are controlled by will. For example, the face consists of many voluntary muscles that facilitate expression.
2. Nonstriated (smooth) or involuntary muscles function without the action of the will. These muscles are found in the internal organs of the body, such as the stomach and intestines.
3. Cardiac muscle is the heart itself. An involuntary muscle with striations and centrally located nuclei, it is not duplicated anywhere else in the body.

### Muscle Structure

When a muscle contracts and shortens, one of its attachments usually remains fixed and the other one moves. A muscle has three parts (Figure 6-8):

- **Origin** (OR-ih-jin). The part of the muscle that does not move and is attached closest to the skeleton.
- **Belly** (BELL-ee). The middle part of the muscle.
- **Insertion** (in-SUR-shun). The part of the muscle that moves and is farthest from the skeleton.

figure 6-6  
Bones of the arm.

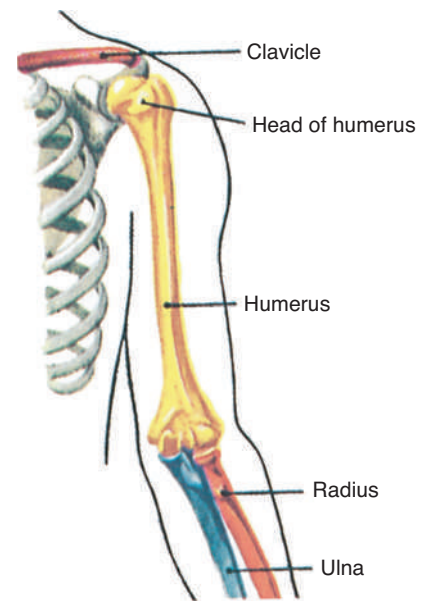
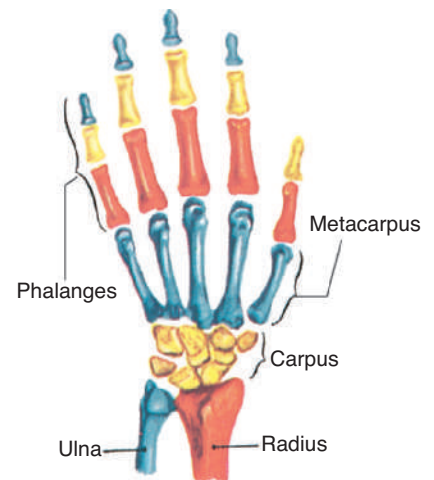
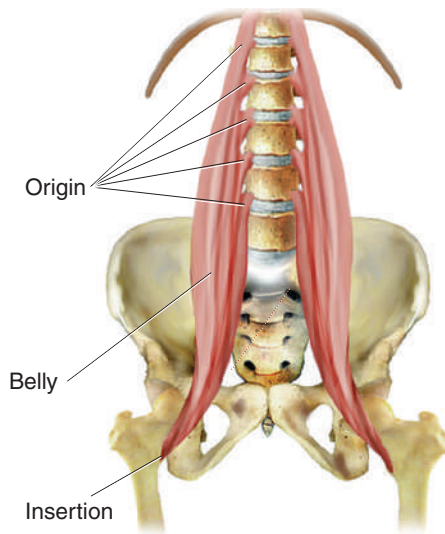


figure 6-7  
Bones of the hand.





**figure 6-8**  
Muscle origin and insertion.



## Stimulation of Muscles

Muscular tissue can be stimulated by:

- Massage (hand, electric vibrator, or water jets).
- Electrical therapy current. (See Chapter 8, Basics of Electricity, for additional information on types of electrical therapy current.)
- Infrared light.
- Dry heat (heating lamps or heating caps).
- Moist heat (steamers or moderately warm steam towels).
- Nerve impulses (through the nervous system).
- Chemicals (certain acids and salts).

## Muscles of the Scalp, Face, and Neck

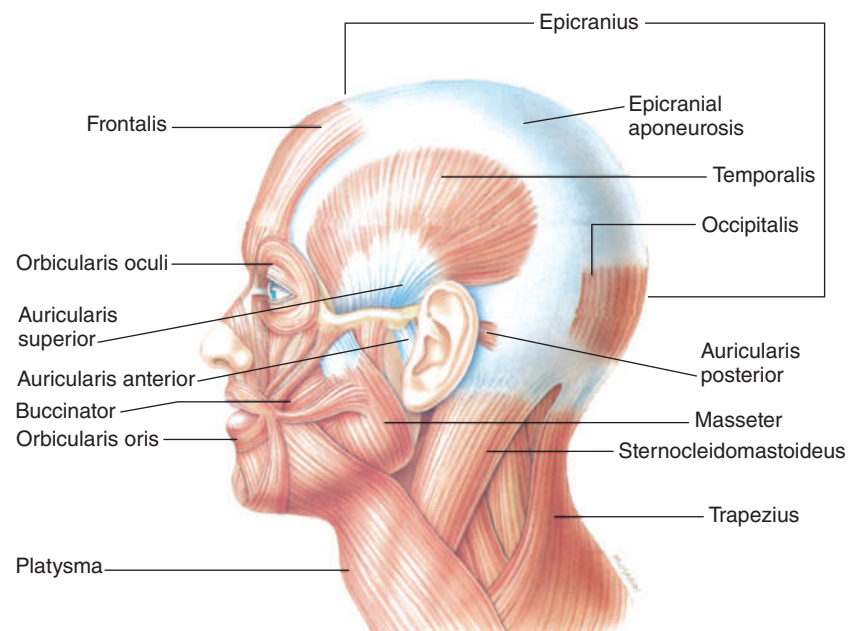
The performance of professional services such as scalp, facial, and neck massages requires barbers to know the location of the voluntary muscles of the head, face, and neck and the functions these muscles control (**Figures 6-9 and 6-10**).

### Muscles of the Scalp

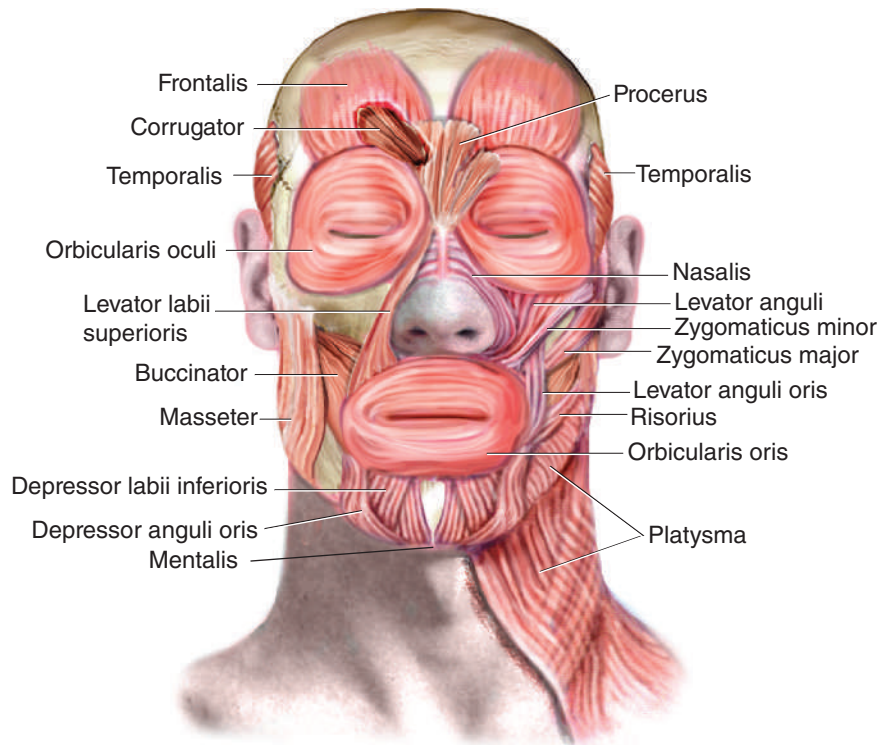
The four muscles of the scalp are the following:

- **Epicranius** (ep-ih-KRAY-nee-us), also known as *occipitofrontalis* (ahk-SIP-ih-toh-frun-TAY-lus). Broad muscle that covers the top of the skull and consists of the occipitalis and frontalis.
- **Occipitalis** (ahk-SIP-ih-tahl-is). Back (posterior) portion of the epicranius; the muscle that draws the scalp backward.
- **Frontalis** (frun-TAY-lus). Front (anterior) portion of the epicranius; the muscle of the scalp that raises the eyebrows, draws the scalp forward, and causes wrinkles across the forehead.
- **Epicranial aponeurosis** (ep-ih-KRAY-nee-al ap-uh-noo-ROH-sus). Tendon that connects the occipitalis and frontalis muscles (**Figure 6-9**).

**figure 6-9**  
Muscles of the head, face, and neck.



**figure 6-10**  
Muscles of the face.



### Muscles of the Ear

The three muscles of the ear are the following:

- **Auricularis superior** (aw-rik-yuh-LAIR-is soo-PEER-ee-ur). Muscle above the ear that draws the ear upward.
- **Auricularis anterior** (aw-rik-yuh-LAIR-is an-TEER-ee-ur). Muscle in front of the ear that draws the ear forward.
- **Auricularis posterior** (aw-rik-yuh-LAIR-is poh-STEER-ee-ur). Muscle behind the ear that draws the ear backward.

### Muscles of Mastication (Chewing)

The main muscles of mastication, also known as the *chewing muscles*, are the following:

- **Masseter** (muh-SEET-ur).
- **Temporalis** (tem-poh-RAY-lis).

These muscles coordinate to open and close the mouth and bring the jaw forward or backward, assisted by the *pterygoid* (THER-ih-goyd) muscles.

### Muscles of the Neck

The two major muscles of the neck are the following:

- The **platysma** (plah-TIZ-muh) is a broad muscle extending from the chest and shoulder muscles to the side of the chin and is responsible for depressing the lower jaw and lip.
- The **sternocleidomastoideus** (STUR-noh-KLEE-ih-doh-mas-TOYD-ee-us) is the muscle extending from the collar and chest bones to the temporal bone at the back of the ear; this muscle bends and rotates the head.

## Muscles of the Eye

The eye muscles include the following:

- **Orbicularis oculi muscle** (or-bik-yuh-LAIR-is AHK-yuh-lye MUS-uhl). Ring muscle of the eye socket; enables you to close your eyes.
- **Corrugator muscle** (KOR-uh-gayt-or MUS-uhl). Muscle located beneath the frontalis and orbicularis oculi muscle that draws the eyebrow down and wrinkles the forehead vertically.
- **Levator palpebrae superioris muscle** (lih-VAYT-ur PAL-puh-bree soo-peer-ee-OR-is MUS-uhl). Thin muscle that controls the eyelid and can be easily damaged during massage or makeup application.

## Muscles of the Nose

- The **procerus** (proh-SEE-rus) covers the top of the nose, depresses the eyebrow, and causes wrinkles across the bridge of the nose. The other nasal muscles are small muscles around the nasal openings, which contract and expand the opening of the nostrils.

## Muscles of the Mouth

The important muscles of the mouth are the following:

- **Buccinator muscle** (BUK-sih-nay-tur MUS-uhl). Thin, flat muscle of the cheek between the upper and lower jaw that compresses the cheeks and expels air between the lips.
- **Depressor labii inferioris muscle** (dee-PRES-ur LAY-bee-eye in-FEER-ee-or-us MUS-uhl), also known as *quadratus labii inferioris muscle* (kwah-DRAY-tus LAY-bee-eye in-FEER-ee-or-us MUS-uhl). Muscle surrounding the lower lip; lowers the lower lip and draws it to one side, as in expressing sarcasm.
- **Levator anguli oris muscle** (lih-VAYT-ur ANG-yoo-ly OH-ris MUS-uhl), also known as *caninus muscle* (kay-NY-nus MUS-uhl). Muscle that raises the angle of the mouth and draws it inward.
- **Levator labii superioris muscle** (lih-VAYT-ur LAY-bee-eye soo-peer-ee-OR-is MUS-uhl), also known as *quadratus labii superioris muscle* (kwah-DRAY-tus LAY-bee-eye soo-peer-ee-OR-is MUS-uhl). Muscle surrounding the upper lip; elevates the upper lip and dilates the nostrils, as in expressing distaste.
- **Mentalis muscle** (men-TAY-lis MUS-uhl). Muscle that elevates the lower lip and raises and wrinkles the skin of the chin.
- **Orbicularis oris muscle** (or-bik-yuh-LAIR-is OH-ris MUS-uhl). Flat band of muscle around the upper and lower lips that compresses, contracts, puckers, and wrinkles the lips.
- **Risorius muscle** (rih-ZOR-ee-us MUS-uhl). Muscle of the mouth that draws the corner of the mouth out and back, as in grinning.
- **Triangularis muscle** (try-ang-gyuh-LAY-rus MUS-uhl). Muscle extending alongside the chin that pulls down the corner of the mouth.

- **Zygomaticus major muscles** (zy-goh-MAT-ih-kus MAY-jor MUS-uhlz). Muscles on both sides of the face that extend from the zygomatic bone to the angle of the mouth. These muscles pull the mouth upward and backward, as when you are laughing or smiling.
- **Zygomaticus minor muscles** (zy-goh-mat-ih-kus MY-nor MUS-uhlz). Muscles on both sides of the face that extend from the zygomatic bone to the upper lips. These muscles pull the upper lip backward, upward, and outward, as when you are smiling.

### **Muscles That Attach the Arms to the Body**

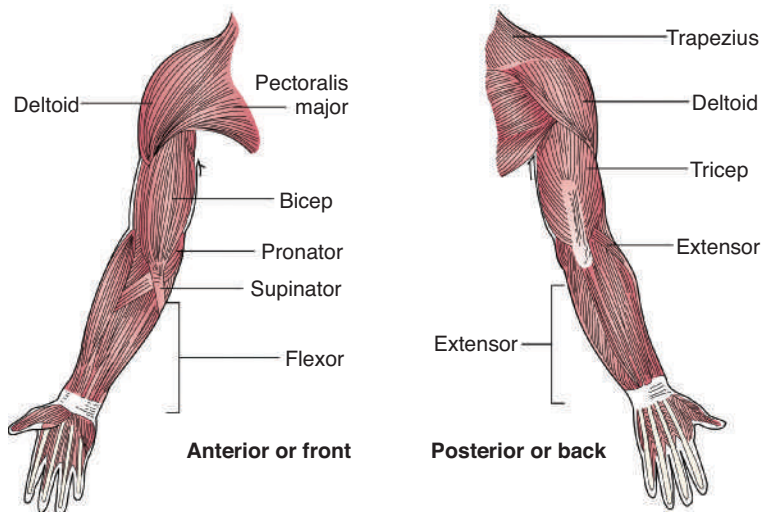
The muscles that attach the arms to the body are the following:

- **Latissimus dorsi** (lah-TIS-ih-mus DOR-see). Large, flat, triangular muscle covering the lower back. It helps extend the arm away from the body and rotate the shoulder.
- **Pectoralis major** (pek-tor-AL-is MAY-jor) and **pectoralis minor** (located under the pectoralis major) are muscles of the chest that assist the swinging movements of the arm.
- **Serratus anterior** (ser-RAT-us an-TEER-ee-ur). Muscle of the chest that assists in breathing and in raising the arm.
- **Trapezius** (truh-PEE-zee-us). Muscle that covers the back of the neck and the upper and middle region of the back; rotates and controls swinging movements of the arm (**Figures 6-11** and **6-12**).

### **Muscles of the Shoulder and Arm**

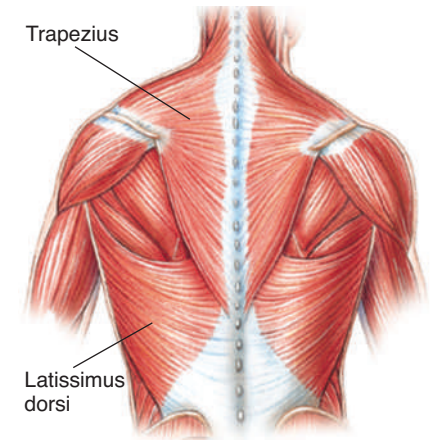
There are three principal muscles of the shoulders and upper arms (**Figure 6-13**):

- **Bicep** (BY-sep). Muscle that produces the contour of the front and inner side of the upper arm; lifts the forearm and flexes the elbow.
- **Deltoid** (DEL-toyd). Large, triangular muscle covering the shoulder joint that allows the arm to extend outward and to the side of the body.



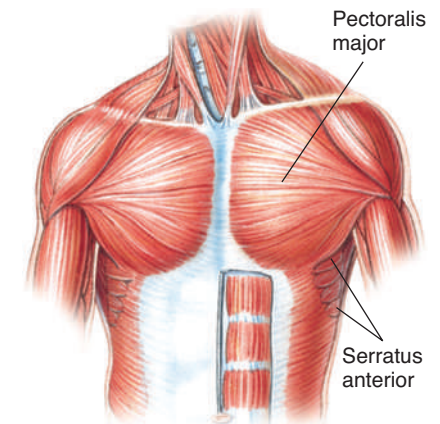
**figure 6-11**

Muscles of the back that attach the arms to the body.



**figure 6-12**

Muscles of the chest that attach the arms to the body.



**figure 6-13**

Muscles of the anterior and posterior shoulder and arm.



- **Tricep** (TRY-sep). Large muscle that covers the entire back of the upper arm and extends the forearm.

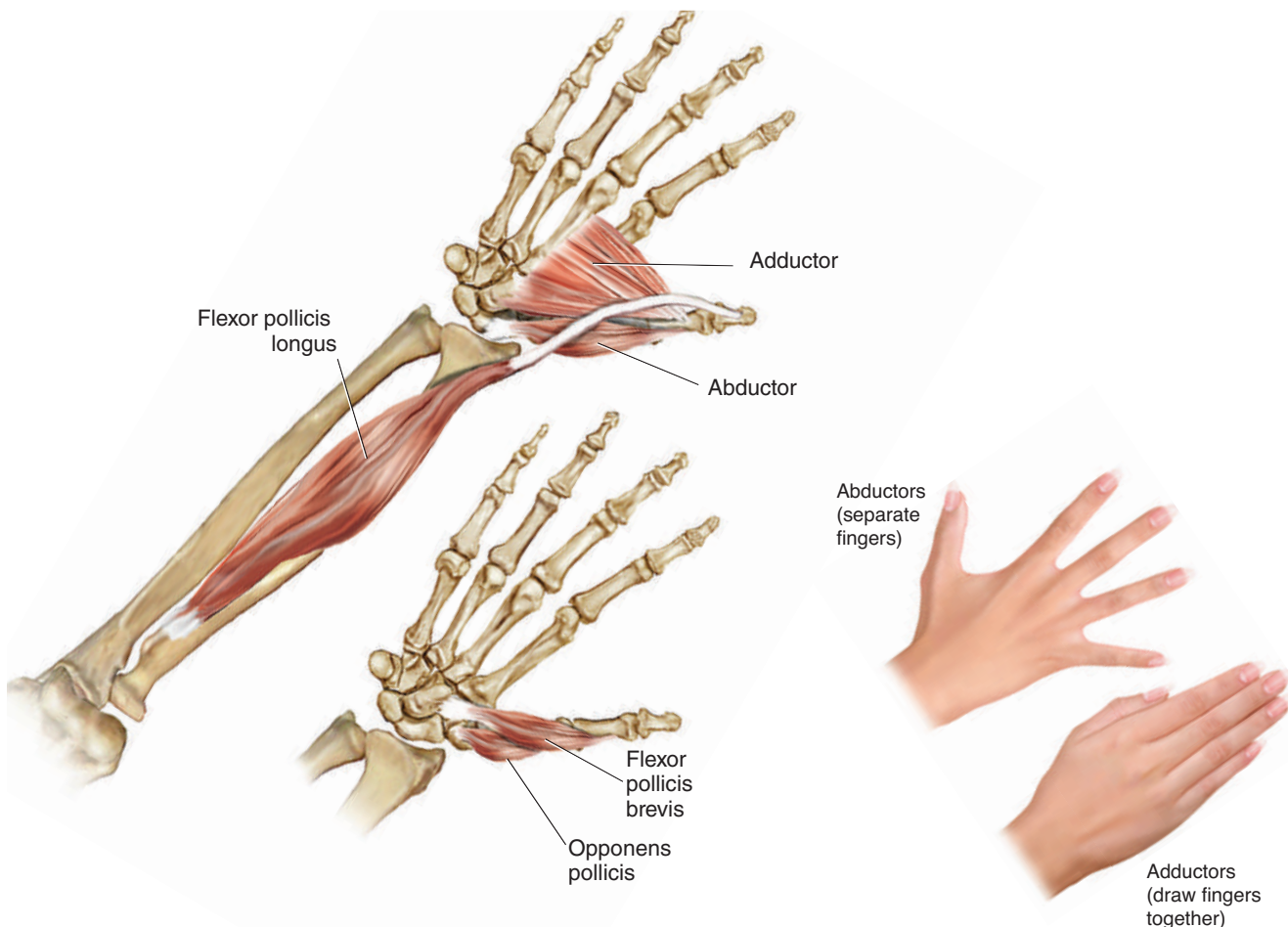
The forearm is made up of a series of muscles and strong tendons. As a barber, you will be concerned with the following muscles of the forearm:

- **Extensors** (ik-STEN-surs). Muscles that straighten the wrist, hand, and fingers to form a straight line.
- **Flexor** (FLEK-sur). Extensor muscle of the wrist involved in flexing the wrist.
- **Pronator** (proh-NAY-tohr). Muscle that turns the hand inward so that the palm faces downward.
- **Supinator** (SOO-puh-nayt-ur). Muscle of the forearm that rotates the radius outward and the palm upward.

### ***Muscles of the Hand***

The hand is one of the most complex parts of the body, with many small muscles that overlap from joint to joint and provide the flexibility and strength to open and close the hand and fingers. Important muscles to know include the following (**Figure 6-14**):

**figure 6-14**  
Muscles of the hand.



- **Abductors** (ab-DUK-turz). Muscles that draw a body part, such as a finger, arm, or toe, away from the midline of the body or of an extremity. In the hand, abductors separate the fingers.
- **Adductors** (ah-DUK-turz). Muscles that draw a body part, such as a finger, arm, or toe, inward toward the median axis of the body or of an extremity. In the hand, adductors draw the fingers together.

## REVIEW THE NERVOUS SYSTEM

*Neurology* is the study of the structure, function, and pathology of the nervous system. The nervous system is one of the most important systems of the body. It controls and coordinates the functions of all the other systems and makes them work harmoniously and efficiently. Every square inch of the human body is supplied with fine fibers known as *nerves*.

An understanding of how nerves work will help barbers perform the massage services associated with shampoos, scalp treatments, and facials. It will also help increase awareness of the effects these treatments can have on the skin, the scalp, and on the body as a whole.

### Divisions of the Nervous System

The principal components of the nervous system are the brain, the spinal cord, and the nerves themselves. The nervous system is divided into two main subdivisions: the central nervous system and the peripheral nervous system (Figure 6-15).

1. The **central nervous system (CNS)** (SEN-trul NUR-vus SIS-tum) consists of the brain, cranial nerves, spinal cord, and spinal nerves. It controls consciousness and all mental activities, the functions of the five senses, and voluntary muscle actions, including all body movements and facial expressions.
2. The **peripheral nervous system (PNS)** (puh-RIF-uh-rul NUR-vus SIS-tum) is made up of sensory and motor nerve fibers that extend from the brain and spinal cord to all parts of the body. Their function is to carry impulses, or messages, to and from the central nervous system. Subdivisions of the peripheral nervous system are the autonomic and somatic nervous systems, which control involuntary and voluntary functions and actions.

### The Brain and Spinal Cord

The **brain** (BRAYN) is the part of the central nervous system contained in the cranium. It is the largest and most complex organization of nerve tissue and controls sensation, muscles, activity of glands, and the power to think, sense, and feel. On average, the brain weighs a little less than three pounds. It sends and receives messages through 12 pairs of cranial nerves that originate in the brain and reach various parts of the head, face, and neck.

The **spinal cord** (SPY-nal KORD) is the portion of the central nervous system that originates in the brain and extends down to the lower extremity of the trunk. It is protected by the spinal column. Thirty-one pairs of spinal nerves extending from the spinal cord are distributed to the muscles and skin of the trunk and limbs.

**figure 6-15**  
Divisions of the nervous system.

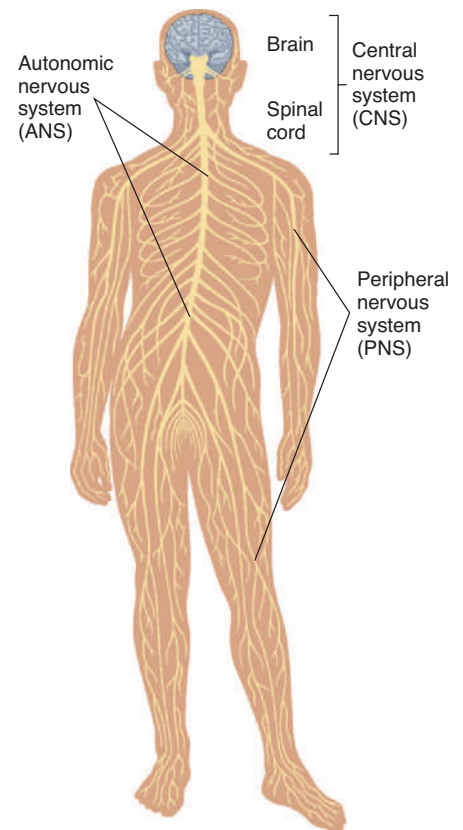
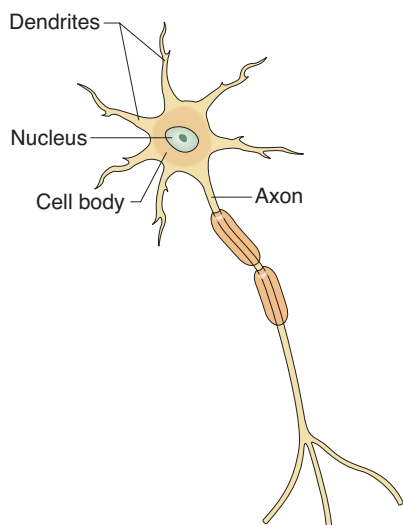


figure 6-16

A neuron or nerve cell.



## Nerve Cell Structure and Function

A **neuron** (NOO-rah-n), also known as *nerve cell*, is the primary structural unit of the nervous system, consisting of the cell body, nucleus, dendrites, and the axon.

**Dendrites** (DEN-dryts) are treelike branchings of nerve fibers extending from the nerve cell that carry impulses toward the cell and receive impulses from other neurons. The **axon** (AK-sahn) and **axon terminal** (AK-sahn TER-min-al) are extensions of a neuron through which impulses are sent away from the cell body to other neurons, glands, or muscles (**Figure 6-16**).

**Nerves** are whitish cords made up of bundles of nerve fibers held together by connective tissue, through which impulses are transmitted. Nerves have their origin in the brain and spinal cord and send their branches to all parts of the body.

## Types of Nerves

There are two types of nerves:

- **Sensory nerves**, also known as *afferent nerves* (AAF-eer-ent NURVZ), carry impulses or messages from the sense organs, where sensations such as touch, cold, heat, sight, hearing, taste, smell, pain, and pressure are experienced. Sensory nerve endings called *receptors* are located close to the surface of the skin. Impulses pass from the sensory nerves to the brain and back through the motor nerves to the muscles; the muscles move as a result of the completed circuit.
- **Motor nerves** (MOH-tur NURVZ), also known as *efferent nerves* (EF-uh-rent NURVZ), carry impulses from the brain to the muscles or glands. These transmitted impulses produce movement.

A **reflex** (REE-fleks) is an automatic reaction to a stimulus that involves the movement of an impulse from a sensory receptor along the sensory nerve to the spinal cord. A responsive impulse is sent along a motor neuron to a muscle, causing a reaction (e.g., the quick removal of your hand from a hot object). Reflexes do not have to be learned; they are automatic.

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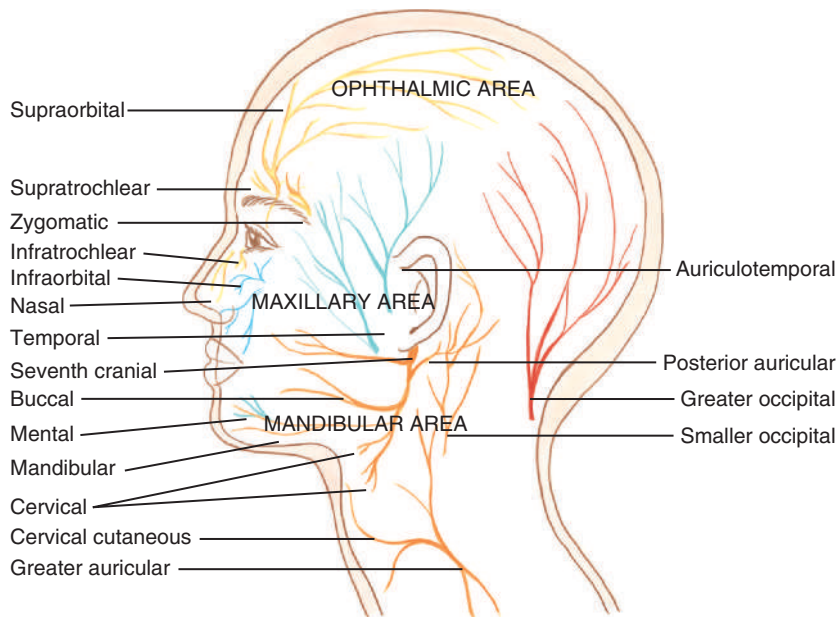
## Activity

There are sensory nerve endings all over the body. Try gently pinching a small piece of skin on your arm. You feel a slight pressure, right? That is the sensory nerve endings sending a message from your arm to your brain that something is happening to your arm.

## Nerves of the Head, Face, and Neck

There are 12 pairs of cranial nerves. All are connected to a part of the brain surface. They emerge through openings on the sides and base of the cranium and reach various parts of the head, face, and neck. The largest of the cranial nerves is the **fifth cranial nerve** (FIFTH KRAY-nee-ul NURV), also known as *trifacial nerve* (try-FAY-shul NURV) or *trigeminal nerve* (try-JEM-un-ul NURV). It is the chief sensory nerve of the face and serves as the motor nerve of the muscles that control chewing. It consists of three branches:

- **Ophthalmic nerve** (ahf-THAL-mik NURV). Supplies impulses to the skin of the forehead, upper eyelids, and interior portion of the scalp, orbit, eyeball, and nasal passage.
- **Mandibular nerve** (man-DIB-yuh-lur NURV). Affects the muscles of the chin, lower lip, and external ear.
- **Maxillary nerve** (MAK-suh-lair-ee NURV). Supplies impulses to the upper part of the face (**Figure 6-17**).



**figure 6-17**  
Nerves of the head, face, and neck.

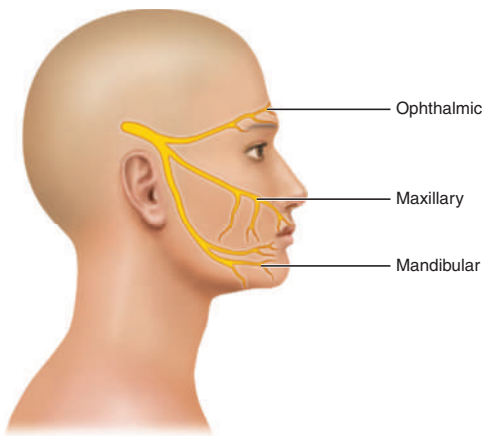
The following are the branches of the fifth cranial nerve that are affected by massage (**Figure 6-18**):

- **Auriculotemporal nerve** (aw-RIK-yuh-loh-TEM-puh-rul NURV). Affects the external ear and skin above the temple, up to the top of the skull.
- **Infraorbital nerve** (in-fruh-OR-bih-tul NURV). Affects the skin of the lower eyelid, side of the nose, upper lip, and mouth.
- **Infratrochlear nerve** (in-frah-TRAHK-lee-ur NURV). Affects the membrane and skin of the nose.
- **Mental nerve** (MEN-tul NURV). Affects the skin of the lower lip and chin.
- **Nasal nerve** (NAY-zul NURV). Affects the tip and lower side of the nose.
- **Supraorbital nerve** (soo-pruh-OR-bih-tul NURV). Affects the skin of the forehead, scalp, eyebrow, and upper eyelid.
- **Supratrochlear nerve** (soo-pruh-TRAHK-lee-ur NURV). Affects the skin between the eyes and upper side of the nose.
- **Zygomatic nerve** (zy-goh-MAT-ik NURV). Affects the muscles of the upper part of the cheek.

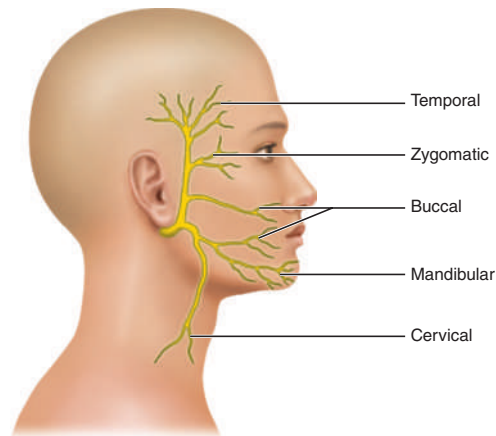
The **seventh cranial nerve** (SEV-AHNTH CRAN-ee-ahl NURV), also known as the *facial nerve*, is the chief motor nerve of the face (**Figure 6-19**). Its divisions



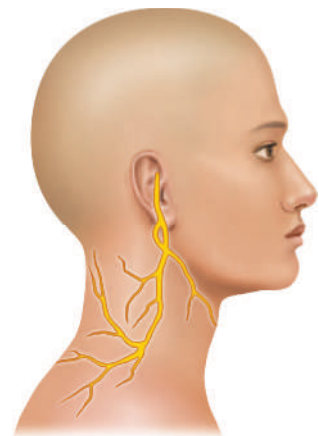
**figure 6-18**  
Fifth cranial nerve.



**figure 6-19**  
Seventh cranial nerve.



**figure 6-20**  
Eleventh cranial nerve.



and their branches supply and control all the muscles of facial expression. It emerges near the lower part of the ear and extends to the muscles of the neck. The following are the most important branches of the facial nerve:

- **Posterior auricular nerve** (poh-STEER-ee-ur aw-rik-yuh-lur NURV). Affects the muscles behind the ear at the base of the skull.
- **Temporal nerve** (TEM-poh-rul NURV). Affects the muscles of the temple, side of the forehead, eyebrow, eyelid, and upper part of the cheek.
- **Zygomatic nerve** (upper and lower). Affects the muscles of the upper part of the cheek.
- **Buccal nerve** (BUK-ul NURV). Affects the muscles of the mouth.
- **Marginal mandibular nerve** (MAR-jin-ul man-DIB-yuh-lur NURV). Affects the muscles of the chin and lower lip.

The **eleventh cranial nerve** (ee-LEV-unth GRAY-nee-ul NURV), also known as *accessory nerve*, is a motor nerve that controls the motion of the neck and shoulder muscles (**Figure 6-20**). This nerve is important to barbers because it is affected during facials, primarily when you are giving a massage to your client.

### **Cervical Nerves**

The cervical or spinal nerves originate at the spinal cord. Their branches supply the muscles and scalp at the back of the head and neck as follows:

- The **greater occipital nerve** (GRAY-tur ahk-SIP-ut-ul NURV) is located at the back of the head and affects the scalp as far up as the top of the head.
- The **smaller (lesser) occipital nerve** (SMAWL-ur ahk-SIP-ut-ul NURV) is located at the base of the skull and affects the scalp and muscles of this region.
- The **greater auricular nerve** (GRAYT-ur aw-RIK-yuh-lur NURV) is located at the side of the neck and affects the external ears and the areas in front and back of the ears.
- The **cervical cutaneous nerve** (SUR-vih-kul kyoo-TAY-nee-us NURV) is located at the side of the neck and affects the front and sides of the neck as far down as the breastbone.

## REVIEW THE CIRCULATORY SYSTEM

The **circulatory system** (SUR-kyool-ah-tohr-ee SIS-tum), also known as *cardio-vascular system* (KAHRD-ee-oh-VAS-kyoo-lur SIS-tum) or *vascular system*, controls the steady circulation of the blood through the body by means of the heart and blood vessels. The circulatory system consists of the heart, arteries, veins, and capillaries that distribute blood throughout the body.

### The Heart

The **heart** (HART) is a muscular, cone-shaped organ that keeps the blood moving within the circulatory system. It is often referred to as the *body's pump*.

The heart is approximately the size of a closed fist, weighs about nine ounces, and is located in the chest cavity. A normal adult heart beats about 60 to 80 times per minute, but it can beat as high as 100 times per minute when necessary and as low as 50 beats per minute in very physically fit individuals. Heart rate norms vary with age and fitness.

The blood is in constant and continuous circulation from the time that it leaves the heart, is distributed throughout the body to deliver nutrients and oxygen, and then returns to the heart to be replenished with oxygen.

Two systems are important to this circulation (**Figure 6-21**):

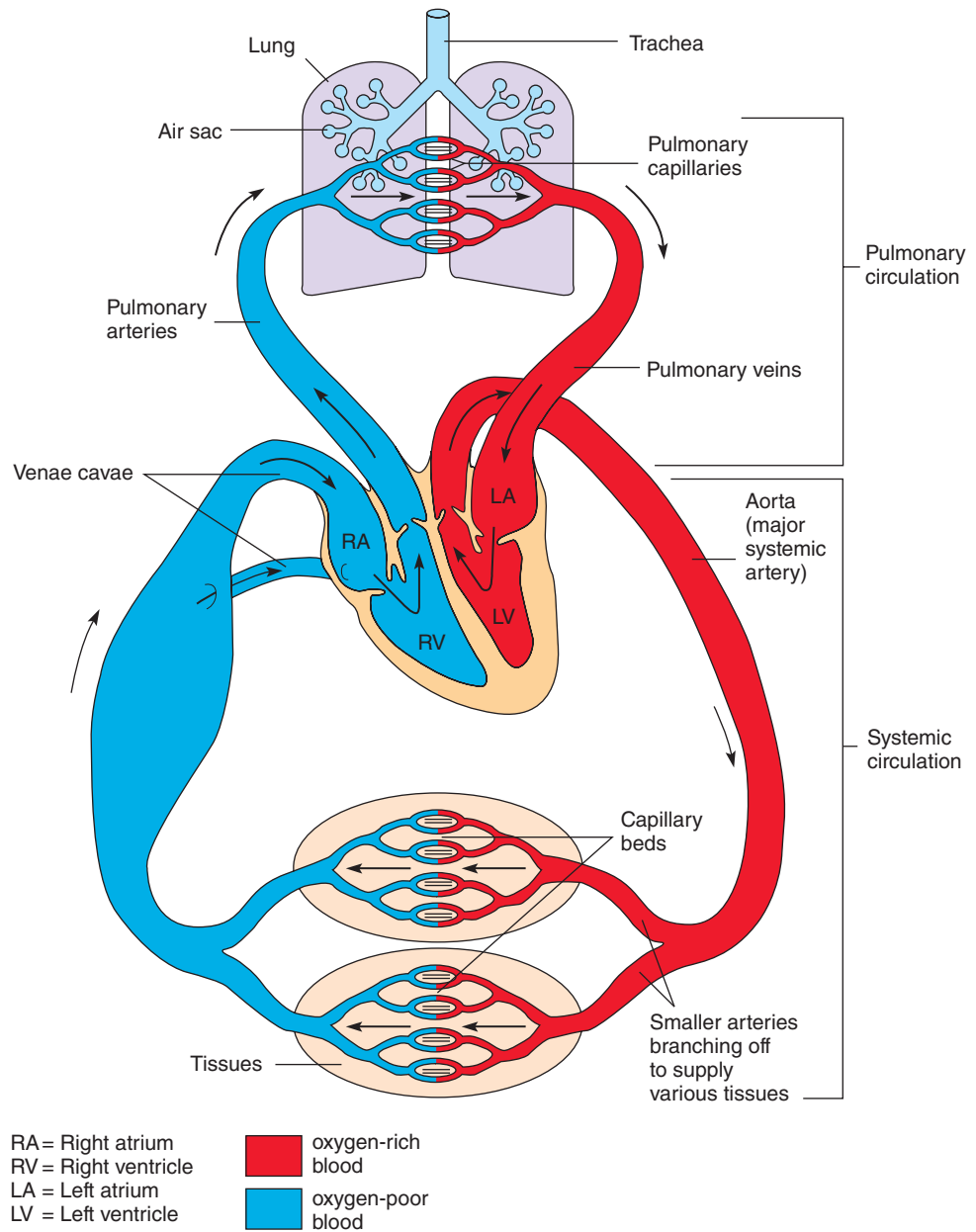
- **Pulmonary circulation** (PUL-muh-nayr-ee sur-kyoo-LAY-shun). Takes deoxygenated blood to the lungs for oxygenation and waste removal and then returns that blood to the heart (left atrium) so oxygen-rich blood can be delivered to the body.
- **Systemic circulation** (sis-TEM-ik sur-kyoo-LAY-shun), also known as **general circulation**. Carries the oxygen-rich blood from the heart throughout the body and returns deoxygenated blood back to the heart.

### Blood Vessels

The **blood vessels** are tubelike structures that include the arteries, arterioles, capillaries, venules, and veins. The function of these vessels is to transport blood to and from the heart and then to various tissues of the body. The types of blood vessels found in the body are:

- **Arteries** (AR-tuh-reez). Thick-walled, muscular, flexible tubes that carry oxygenated blood away from the heart to the arterioles. The largest artery in the body is the **aorta** (ay-ORT-uh).
- **Arterioles** (ar-TEER-ee-ohls). Small arteries that deliver blood to capillaries.
- **Capillaries** (KAP-ih-lair-eez). Tiny, thin-walled blood vessels that connect the smaller arteries to venules. Capillaries bring nutrients to the cells and carry away waste materials.
- **Venules** (VEEN-yools). Small vessels that connect the capillaries to the veins. They collect blood from the capillaries and drain it into the veins.
- **Veins** (VAYNS). Thin-walled blood vessels that are less elastic than arteries; veins contain cuplike valves that keep blood flowing in one direction to the heart and prevent blood from flowing backward. Veins carry blood containing waste products back to the heart and lungs for cleaning and to pick up oxygen. Veins are located closer to the outer skin surface of the body than arteries.

**figure 6-21**  
Drawing of blood flow through the heart.



## Blood

**Blood** (BLUD) is a nutritive fluid circulating through the circulatory system (heart and blood vessels) to supply oxygen and nutrients to cells and tissues and to remove carbon dioxide and waste from them. There are approximately 8 to 10 pints of blood in the human body, which contribute about  $\frac{1}{20}$ th of the body's weight. Blood is approximately 80 percent water. It is sticky and salty, with a normal temperature of 98.6 degrees Fahrenheit. It is bright red in the arteries (except for the pulmonary artery) and dark red in the veins. The color change occurs with the exchange of carbon dioxide for oxygen as the blood passes through the lungs, and again with the exchange of oxygen for carbon dioxide as the blood circulates throughout the body.

## Chief Functions of the Blood

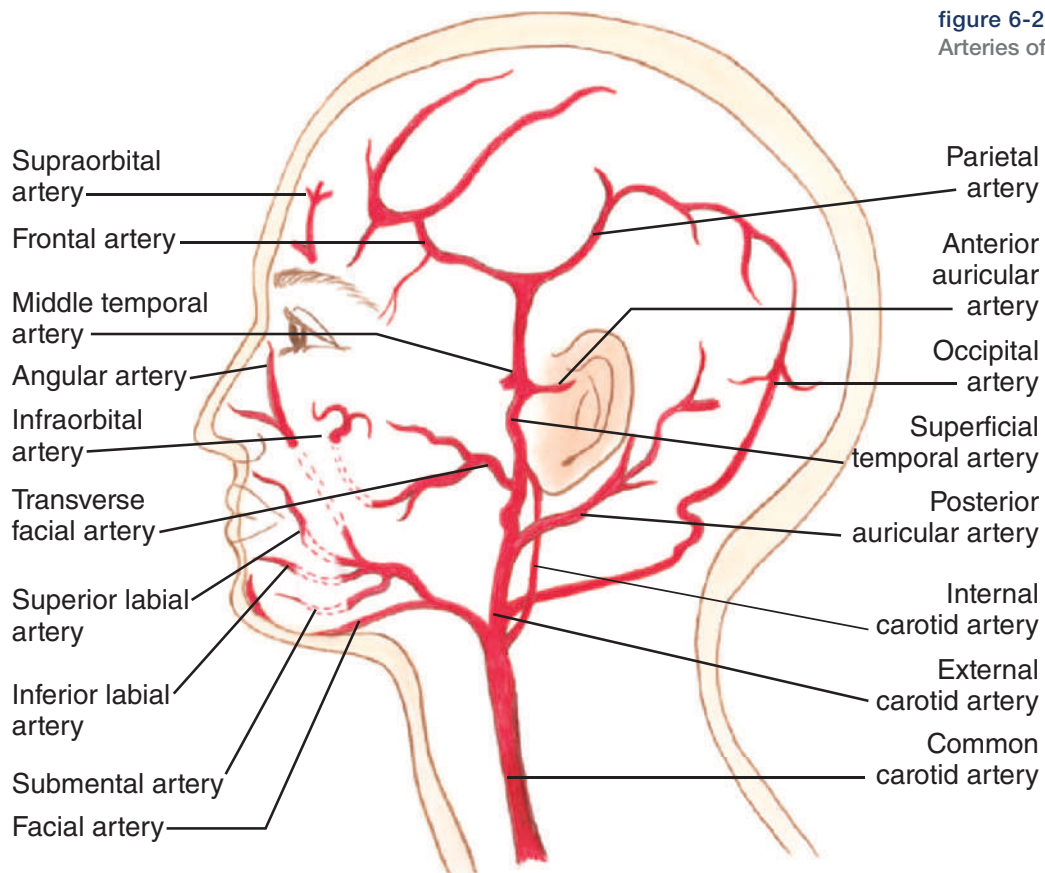
Blood performs the following critical functions:

- Carries water, oxygen, and food to all cells and tissues of the body
- Carries away carbon dioxide and waste products to be eliminated through the lungs, skin, kidneys, and large intestines
- Helps equalize the body's temperature, thus protecting the body from extreme heat and cold
- Works with the immune system to protect the body from harmful toxins and bacteria
- Seals leaks found in injured blood vessels by forming clots, thus preventing further blood loss

## Arteries of the Head, Face, and Neck

The **common carotid arteries** (KAHM-un kuh-RAHT-ud ART-uh-rees) are the main arteries that supply blood to the head, face, and neck. They are located on both sides of the neck, and each artery is divided into an internal and external branch.

The **internal carotid artery** (in-TUR-nul kuh-RAHT-ud ART-uh-ree) supplies blood to the brain, eyes, eyelids, forehead, nose, and internal ear. The **external carotid artery** supplies blood to the anterior (front) parts of the scalp, ears, face, neck, and sides of the head (**Figure 6-22**).



**figure 6-22**  
Arteries of the head, face, and neck.



Two branches of the internal carotid artery that are important to know are the following:

- **Supraorbital artery** (soo-pruh-OR-bih-tul AR-tuh-ree). Supplies blood to the upper eyelid and forehead.
- **Infraorbital artery** (in-fruh-OR-bih-tul ARE-ter-ee). Supplies blood to the muscles of the eye.

There are four branches of the external carotid artery—the facial artery, the superficial temporal artery, the occipital artery, and the posterior auricular artery.

The **facial artery** (FAY-shul ART-uh-ree), also known as the *external maxillary artery* (eks-TUR-nul MAK-sah-lair-ee ART-uh-ree), supplies blood to the lower region of the face, mouth, and nose. Some of the important facial artery branches include:

- **Submental artery** (sub-MEN-tul AR-tuh-ree). Supplies blood to the chin and lower lip.
- **Inferior labial artery** (in-FEER-ee-ur LAY-bee-al ART-ur-ee). Supplies blood to the lower lip.
- **Angular artery** (ANG-yoo-lur ART-ur-ee). Supplies blood to the side of the nose.
- **Superior labial artery** (soo-PEER-ee-ur LAY-bee-ul AR-tuh-ree). Supplies blood to the upper lip and region of the nose.

The **superficial temporal artery** (soo-pur-FISH-ul TEM-puh-rul AR-tuh-ree) is a continuation of the external carotid artery and supplies blood to the muscles of the front, side, and top of the head. Some of the important superficial temporal artery branches include:

- **Frontal artery** (FRUNT-ul ART-uh-ree). Supplies blood to the forehead and upper eyelids.
- **Parietal artery** (puh-RY-ate-ul ART-uh-ree). Supplies blood to the side and crown of the head.
- **Transverse facial artery** (tranz-VURS FAY-shul AR-tuh-ree). Supplies blood to the skin and masseter muscle.
- **Middle temporal artery** (MID-ul TEM-puh-rul ART-uh-ree). Supplies blood to the temples.
- **Anterior auricular artery** (an-TEER-ee-ur aw-RIK-yuh-lur ART-ur-ee). Supplies blood to the front part of the ear.

The **occipital artery** (ahk-SIP-it-ul AR-tuh-ree) supplies blood to the skin and muscles of the scalp and back of the head up to the crown.

The **posterior auricular artery** (poh-STEER-ee-ur aw-RIK-yuh-lur AR-tuh-ree) supplies blood to the scalp, the area behind and above the ear, and the skin behind the ear.

### ***Veins of the Head, Face, and Neck***

The blood returning to the heart from the head, face, and neck flows on each side of the neck in two principal veins:

- The **internal jugular vein** (in-TUR-nul JUG-yuh-lur VAYN) is located at the side of the neck to collect blood from the brain and parts of the face and neck.
- The **external jugular vein** (eks-TUR-nul JUG-yuh-lur VAYN) is located at the side of the neck and carries blood returning to the heart from the head, face, and neck.

The most important veins of the face and neck are parallel to the arteries and take the same names as the arteries.

## REVIEW THE LYMPHATIC/IMMUNE SYSTEM

The **lymphatic/immune system** (lim-FAT-ik/ih-MYOON SIS-tum) is made up of lymph, lymph nodes, the thymus gland, the spleen, and lymph vessels. The lymphatic/immune system carries waste and impurities away from the cells and protects the body from disease by developing immunity and destroying disease-causing microorganisms. **Lymph** (LIMF) is a clear fluid that circulates in the lymph spaces (lymphatics) of the body. Lymph helps carry wastes and impurities away from the cells before it is routed back to the circulatory system. The lymphatic/immune system drains the tissue spaces of excess **interstitial fluid** (in-tur-STISH-al FLOO-id), which is blood plasma found in the spaces between tissue cells. The lymphatic/immune system is closely connected to the cardiovascular system. They both transport streams of fluids, like rivers throughout the body. The difference is that the lymphatic/immune system transports lymph, which eventually returns to the blood where it originated.

Lymphatic vessels start as tubes that are closed at one end. They can occur individually or in clusters that are called **lymph capillaries**, blind-end tubes that are the origin of lymphatic vessels. The lymph capillaries are distributed throughout most of the body (except the nervous system).

**Lymph nodes** are glandlike structures found inside lymphatic vessels. Lymph nodes filter the lymphatic vessels, which helps fight infection.

The primary functions of the lymphatic/immune system are to:

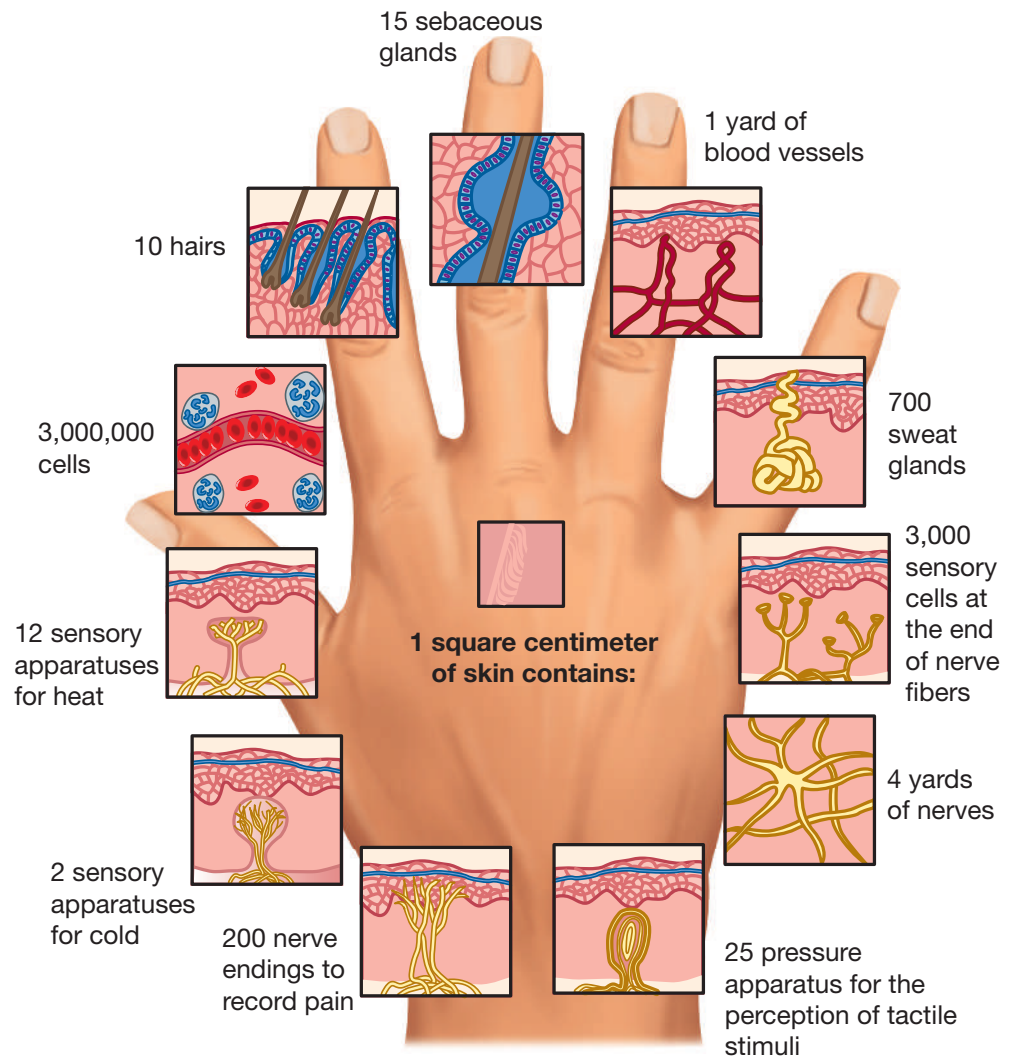
- Carry nourishment from the blood to the body cells.
- Act as a defense against toxins and bacteria, and remove by-products of infection such as pus and dead tissue.
- Remove waste material from the body cells to the blood.
- Provide a suitable fluid environment for the cells.

## REVIEW THE INTEGUMENTARY SYSTEM

The **integumentary system** (in-TEG-yuh-ment-uh-ree SIS-tum) consists of the skin and its accessory organs, such as the oil and sweat glands, sensory receptors, hair, and nails. It is a very complex system that serves as a protective covering and helps regulate the body's temperature (**Figure 6-23**).

The word *integument* means “a natural covering.” So you can think of the skin as a protective overcoat for your body against the outside elements that you encounter every day, such as germs, chemicals, and sun exposure. Skin is also water resistant.

**figure 6-23**  
Structures of the skin.



Skin structure and growth are discussed in detail in Chapter 9, The Skin—Structures, Disorders, and Diseases.

## REVIEW THE ENDOCRINE SYSTEM

The **endocrine system** (EN-duh-krin SIS-tum) is a group of specialized glands that affect the growth, development, sexual functions, and health of the entire body. **Glands** are secretory organs that remove and release certain elements from the blood to convert them into new compounds (Figure 6-24).

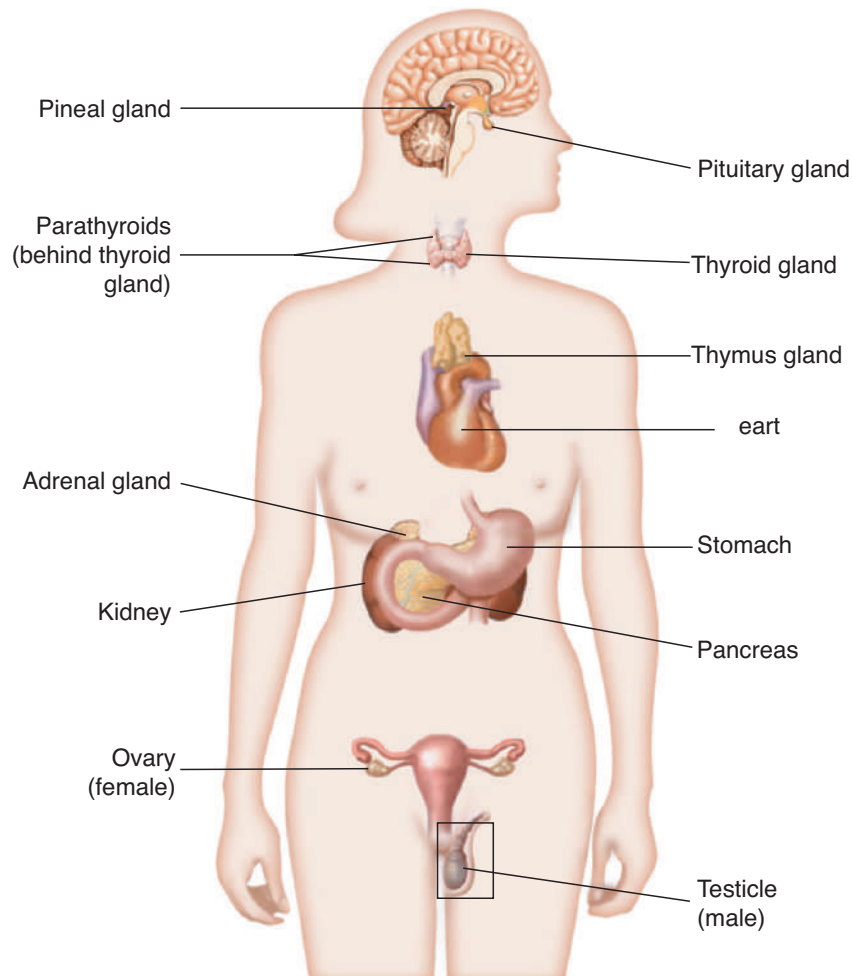
There are two main types of glands:

- **Endocrine glands**, also known as *ductless glands*, such as the thyroid and pituitary glands, release hormonal secretions directly into the bloodstream.
- **Exocrine glands** (EK-suh-krin GLANDZ), also known as *duct glands*, such as sweat and oil glands of the skin, produce a substance that travels through small, tubelike ducts.

### DID YOU KNOW?

Every minute, you shed about 30,000 to 40,000 dead skin cells from your body. That can total up to about 40 pounds of skin in your lifetime!

- **Hormones** (HOR-mohnz) are secretions, such as insulin, adrenaline, and estrogen, that stimulate functional activity or other secretions in the body. Hormones influence the welfare of the entire body.



**figure 6-24**

Endocrine glands and other body organs.

## REVIEW THE REPRODUCTIVE SYSTEM

The **reproductive system** (ree-proh-DUK-tiv SIS-tum) includes the ovaries, uterine tubes, uterus, and vagina in the female and the testes, prostate gland, penis, and urethra in the male. This system performs the function of producing offspring and passing on the genetic code from one generation to another.

The reproductive system produces hormones—primarily estrogen in females and primarily testosterone in males. These hormones affect and change the skin in several ways. Acne, loss of scalp hair, facial hair growth and color, and darker skin pigmentations are some of the results of changing or fluctuating hormones.

## REVIEW QUESTIONS

- 1 Define *anatomy*, *physiology*, and *histology*.
- 2 Identify the anatomical parts of the body that barbers will be most concerned with.
- 3 What is a cell?
- 4 Define *mitosis*.
- 5 Describe the composition of body tissues.
- 6 What is an *organ*?
- 7 List eight important organs of the body.
- 8 What are *systems*?
- 9 Name the 11 body systems and their main functions.
- 10 List the primary functions of the skeletal system.
- 11 Name the two parts of the skull and the number of bones in each.
- 12 List the bones in the cranium.
- 13 Identify the bone that joins all the cranial bones together.
- 14 List the bones in the facial skeleton with their locations and functions.
- 15 Where is the *hyoid bone* located?
- 16 Describe the function and composition of muscles.
- 17 List and describe the parts of a muscle.
- 18 List the ways in which muscles may be stimulated.
- 19 List the scalp, facial, and neck muscles with their locations and functions.
- 20 List and describe the functions of the two main divisions of the nervous system.
- 21 List and describe the two types of nerves found in the human body.
- 22 List the ways in which nerves may be stimulated.
- 23 Give an example of nerve reflex.
- 24 Name the cranial nerves most important in massage services and their functions.
- 25 What is the function of the heart? What does it look like?
- 26 List three kinds of vessels found in the circulatory system.
- 27 Identify and describe the two main types of glands in the human body.
- 28 What structures (appendages) are included in the integumentary system?

## CHAPTER GLOSSARY

|                                                                              |        |                                                                                                                            |
|------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------|
| <b>abductors</b> (ab-DUK-turz)                                               | p. 157 | muscles that draw a body part, such as a finger, arm, or toe, away from the midline of the body or of an extremity         |
| <b>adductors</b> (ah-DUK-turz)                                               | p. 157 | muscles that draw a body part, such as a finger, arm, or toe, inward toward the median axis of the body or of an extremity |
| <b>adipose tissue</b><br>(ADD-ih-pohz TISH-oo)                               | p. 147 | a technical term for fat; gives smoothness and contour to the body                                                         |
| <b>anatomy</b> (ah-NAT-ah-mee)                                               | p. 144 | the science of the structure of organisms and of their parts                                                               |
| <b>angular artery</b> (ANG-yoo-lur ART-ur-ee)                                | p. 164 | artery that supplies blood to the sides of the nose                                                                        |
| <b>anterior auricular artery</b><br>(an-TEER-ee-ur aw-RIK-yuh-lur ART-ur-ee) | p. 164 | artery that supplies blood to the front part of the ear                                                                    |
| <b>aorta</b> (ay-ORT-uh)                                                     | p. 161 | largest artery in the body                                                                                                 |



|                                                                     |        |                                                                                                                                                                                          |
|---------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>arteries</b> (AR-tuh-rees)                                       | p. 161 | thick-walled, muscular, flexible tubes that carry oxygenated blood away from the heart to the arterioles                                                                                 |
| <b>arterioles</b> (ar-TEER-ee-ohls)                                 | p. 161 | small arteries that deliver blood to capillaries                                                                                                                                         |
| <b>auricularis anterior</b> (aw-rik-yuh-LAIR-is an-TEER-ee-ur)      | p. 153 | muscle in front of the ear that draws the ear forward                                                                                                                                    |
| <b>auricularis posterior</b> (aw-rik-yuh-LAIR-is poh-STEER-ee-ur)   | p. 153 | muscle behind the ear that draws the ear backward                                                                                                                                        |
| <b>auricularis superior</b> (aw-rik-yuh-LAIR-is soo-PEER-ee-ur)     | p. 153 | muscle above the ear that draws the ear upward                                                                                                                                           |
| <b>auriculotemporal nerve</b> (aw-RIK-yuh-loh-TEM-puh-rul NURV)     | p. 159 | nerve that affects the external ear and skin above the temple, up to the top of the skull                                                                                                |
| <b>axon</b> (AK-sahn) and <b>axon terminal</b> (AK-sahn TER-min-al) | p. 158 | extensions of a neuron through which impulses are sent away from the cell body to other neurons, glands, or muscles                                                                      |
| <b>belly</b> (BELL-ee)                                              | p. 151 | middle part of a muscle                                                                                                                                                                  |
| <b>bicep</b> (BY-sep)                                               | p. 155 | muscle that produces the contour of the front and inner side of the upper arm; lifts the forearm and flexes the elbow                                                                    |
| <b>blood</b> (BLUD)                                                 | p. 162 | nutritive fluid circulating through the circulatory system that supplies oxygen and nutrients to cells and tissues and removes carbon dioxide and waste from them                        |
| <b>blood vessels</b>                                                | p. 161 | tubelike structures that include the arteries, arterioles, capillaries, venules, and veins; function is to transport blood to and from the heart and then to various tissues of the body |
| <b>body systems</b> (BAHD-ee SIS-tums)                              | p. 147 | groups of body organs acting together to perform one or more functions                                                                                                                   |
| <b>brain</b> (BRAYN)                                                | p. 157 | largest and most complex nerve tissue; part of the central nervous system contained within the cranium                                                                                   |
| <b>buccal nerve</b> (BUK-ul NURV)                                   | p. 160 | nerve that affects the muscles of the mouth                                                                                                                                              |
| <b>buccinator muscle</b> (BUK-sih-nay-tur MUS-uhl)                  | p. 154 | thin, flat muscle of the cheek between the upper and lower jaws                                                                                                                          |
| <b>capillaries</b> (KAP-ih-lair-eez)                                | p. 161 | tiny, thin-walled blood vessels that connect the smaller arteries to venules and bring nutrients to the cells and carry away waste materials                                             |
| <b>carpus</b> (KAR-pus)                                             | p. 151 | the bones of the wrist                                                                                                                                                                   |
| <b>cell membrane</b> (SELL mem-brain)                               | p. 146 | part of the cell that encloses the protoplasm; permits soluble substances to enter and leave the cell                                                                                    |
| <b>cells</b> (SELLZ)                                                | p. 145 | basic units of all living things                                                                                                                                                         |
| <b>central nervous system (CNS)</b> (SEN-trul NUR-vus SIS-tum)      | p. 157 | division of nervous system consisting of the brain, spinal cord, spinal nerves, and cranial nerves                                                                                       |
| <b>centrioles</b> (SEN-tree-olz)                                    | p. 146 | two small structures near the nucleus that move to each side during the mitosis process to help divide the cell                                                                          |
| <b>cervical cutaneous nerve</b> (SUR-vih-kul kyoo-TAY-nee-us NURV)  | p. 160 | nerve located at the side of the neck; affects the front and sides of the neck to the breastbone                                                                                         |
| <b>cervical vertebrae</b> (SUR-vih-kul VURT-uh-bray)                | p. 150 | seven bones that form the top part of the spinal column in the neck region                                                                                                               |

|                                                                                             |        |                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>circulatory system</b> (SUR-kyoo-lah-tohr-ee SIS-tum)                                    | p. 161 | system that controls the steady circulation of blood through the body by means of the heart and blood vessels                                                                    |
| <b>common carotid arteries</b> (KAHM-un kuh-RAHT-ud ART-uh-rees)                            | p. 163 | arteries that supply blood to the head, face, and neck                                                                                                                           |
| <b>connective tissue</b> (kuh-NEK-tiv TISH-oo)                                              | p. 147 | fibrous tissue that binds together, protects, and supports the various parts of the body; examples include bone, cartilage, ligaments, tendons, blood, lymph, and adipose tissue |
| <b>corrugator muscle</b> (KOR-uh-gayt-or MUS-uhl)                                           | p. 154 | facial muscle that draws eyebrows down and wrinkles the forehead vertically                                                                                                      |
| <b>cranium</b> (KRAY-nee-um)                                                                | p. 149 | an oval, bony case that protects the brain                                                                                                                                       |
| <b>cytoplasm</b> (sy-toh-PLAZ-um)                                                           | p. 145 | is the watery fluid that surrounds the nucleus of the cell and is needed for growth, reproduction and self-repair.                                                               |
| <b>deltoid</b> (DEL-toyd)                                                                   | p. 155 | a large, triangular muscle covering the shoulder joint that allows the arm to extend outward and to the side of the body                                                         |
| <b>dendrites</b> (DEN-dryts)                                                                | p. 158 | treelike branchings of nerve fibers extending from the nerve cell that carry impulses toward the cell and receive impulses from other neurons                                    |
| <b>depressor labii inferioris muscle</b> (dee-PRES-ur LAY-bee-eye in-FEER-ee-or-us MUS-uhl) | p. 154 | muscle surrounding the lower lip                                                                                                                                                 |
| <b>digestive system</b> (dy-JES-tiv SIS-tum)                                                | p. 148 | the mouth, stomach, intestines, and salivary and gastric glands that change food into nutrients and wastes                                                                       |
| <b>eleventh cranial nerve</b> (ee-LEV-unth CRAY-nee-ul NURV)                                | p. 160 | spinal nerve branch that affects the muscles of the neck and back                                                                                                                |
| <b>endocrine glands</b>                                                                     | p. 166 | also known as <i>ductless glands</i> ; release hormonal secretions directly into the bloodstream, for example, thyroid and pituitary glands                                      |
| <b>endocrine system</b> (EN-duh-krin SIS-tum)                                               | p. 166 | group of specialized glands that affect growth, development, sexual function, and general health                                                                                 |
| <b>epicranial aponeurosis</b> (ep-ih-KRAY-nee-al ap-uh-noo-ROH-sus)                         | p. 152 | a tendon that connects the occipitalis and frontalis muscles                                                                                                                     |
| <b>epicranium</b> (ep-ih-KRAY-nee-us)                                                       | p. 152 | also known as <i>occipitofrontalis</i> (ahk-SIP-ih-toh-frun-TAY-lus) broad muscle that covers the top of the skull and consists of the occipitalis and frontalis                 |
| <b>epithelial tissue</b> (ep-ih-THEE-lee-ul TISH-oo)                                        | p. 147 | a protective covering on body surfaces, such as skin, mucous membranes, the tissue inside the mouth, the lining of the heart, digestive and respiratory organs, and the glands   |
| <b>ethmoid bone</b> (ETH-moyd BOHN)                                                         | p. 149 | a light, spongy bone between the eye sockets forming part of the nasal cavities                                                                                                  |
| <b>excretory system</b> (EKS-kruh-toh-ree SIS-tum)                                          | p. 148 | group of organs including the kidneys, liver, skin, large intestine, and lungs that purify the body by the elimination of waste matter                                           |
| <b>exocrine glands</b> (EK-suh-krin GLANDZ)                                                 | p. 166 | also known as <i>duct glands</i> ; produce a substance that travels through small, tubelike ducts; for example, sweat and oil glands of the skin                                 |
| <b>extensors</b> (ik-STEN-surs)                                                             | p. 156 | muscles that straighten the wrist, hand, and fingers to form a straight line                                                                                                     |
| <b>external carotid artery</b>                                                              | p. 163 | artery that supplies blood to the anterior parts of the scalp, face, neck, and side of the head                                                                                  |

|                                                                    |        |                                                                                                                                                        |
|--------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>external jugular vein</b> (eks-TUR-nul JUG-yuh-lur VAYN)        | p. 165 | vein located at the side of the neck that carries blood returning to the heart from the head, face, and neck                                           |
| <b>facial artery</b> (FAY-shul ART-uh-ree)                         | p. 164 | artery that supplies blood to the lower region of the face, mouth, and nose                                                                            |
| <b>facial skeleton</b> (FAY-shul skell-uh-tin)                     | p. 149 | two nasal bones; two lacrimal bones; two zygomatic bones; two maxillae; the mandible; two turbinal bones; two palatine bones; and the vomer            |
| <b>fifth cranial nerve</b> (FIFTH KRAY-nee-ul NURV)                | p. 158 | chief sensory nerve of the face; controls chewing                                                                                                      |
| <b>flexor</b> (FLEK-sur)                                           | p. 156 | extensor muscle of the wrist involved in flexing the wrist                                                                                             |
| <b>frontal artery</b> (FRUNT-ul ART-uh-ree)                        | p. 164 | artery that supplies blood to the forehead and upper eyelids                                                                                           |
| <b>frontal bone</b> (FRUNT-ul BOHN)                                | p. 149 | bone that forms the forehead                                                                                                                           |
| <b>frontalis</b> (frun-TAY-lus)                                    | p. 152 | anterior or front portion of the epicranium; muscle of the scalp                                                                                       |
| <b>general circulation</b>                                         | p. 161 | see systemic circulation                                                                                                                               |
| <b>glands</b>                                                      | p. 166 | specialized organs varying in size and function that have the ability to remove certain elements from the blood and to convert them into new compounds |
| <b>greater auricular nerve</b> (GRAYT-ur aw-RIK-yuh-lur NURV)      | p. 160 | nerve at the sides of the neck affecting the face, ears, and neck                                                                                      |
| <b>greater occipital nerve</b> (GRAY-tur ahk-SIP-ut-ul NURV)       | p. 160 | nerve located at the back of the head, affecting the scalp                                                                                             |
| <b>heart</b> (HART)                                                | p. 161 | muscular, cone-shaped organ that keeps blood moving through the circulatory system                                                                     |
| <b>histology</b> (his-TAHL-uh-jee)                                 | p. 145 | the study of the minute structure of the various tissues and organs that make up the entire body of an organism                                        |
| <b>hormones</b> (HOR-mohnz)                                        | p. 167 | secretions that stimulate functional activity or other secretions in the body, for example, insulin, adrenaline, and estrogen                          |
| <b>humerus</b> (HYOO-muh-rus)                                      | p. 151 | uppermost and largest bone in the arm                                                                                                                  |
| <b>hyoid bone</b> (HY-oyd BOHN)                                    | p. 150 | U-shaped bone at the base of the tongue at the front part of the throat                                                                                |
| <b>inferior labial artery</b> (in-FEER-ee-ur LAY-bee-al ART-ur-ee) | p. 164 | artery that supplies blood to the lower lip                                                                                                            |
| <b>infraorbital artery</b> (in-fruh-OR-bih-tul ARE-ter-ee)         | p. 164 | artery that supplies blood to the eye muscles                                                                                                          |
| <b>infraorbital nerve</b> (in-fruh-OR-bih-tul NURV)                | p. 159 | nerve that affects the skin of the lower eyelid, side of the nose, upper lip, and mouth                                                                |
| <b>infratrochlear nerve</b> (in-frah-TRAHK-lee-ur NURV)            | p. 159 | nerve that affects the membrane and skin of the nose                                                                                                   |
| <b>insertion</b> (in-SUR-shun)                                     | p. 151 | the part of the muscle that moves and is furthest from the skeleton                                                                                    |
| <b>integumentary system</b> (in-TEG-yuh-ment-uh-ree SIS-tum)       | p. 165 | the skin and its appendages—glands, sensory receptors, nails and hair                                                                                  |
| <b>internal carotid artery</b> (in-TUR-nul kuh-RAHT-ud ART-uh-ree) | p. 163 | artery that supplies blood to the brain, eyes, eyelids, forehead, nose, and ear                                                                        |

|                                                                                                 |        |                                                                                                                           |
|-------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------|
| <b>internal jugular vein</b> (in-TUR-nul JUG-yuh-lur VAYN)                                      | p. 165 | vein located at the side of the neck; collects blood from the brain and parts of the face and neck                        |
| <b>interstitial fluid</b> (in-tur-STISH-al FLOO-id)                                             | p. 165 | blood plasma found in the spaces between tissue cells                                                                     |
| <b>joint</b>                                                                                    | p. 149 | the connection between two or more bones of the skeleton                                                                  |
| <b>lacrimal bones</b> (LAK-ruh-mul BONZ)                                                        | p. 150 | small bones located in the wall of the eye sockets                                                                        |
| <b>latissimus dorsi</b> (lah-TIS-ih-mus DOR-see)                                                | p. 155 | a large, flat, triangular muscle covering the lower back; helps extend the arm away from the body and rotate the shoulder |
| <b>levator anguli oris muscle</b> (lih-VAYT-ur ANG-yoo-ly OH-ris MUS-uhl)                       | p. 154 | muscle that raises the angle of the mouth and draws it inward                                                             |
| <b>levator labii superioris muscle</b> (lih-VAYT-ur LAY-bee-eye soo-peer-ee-OR-is MUS-uhl)      | p. 154 | muscle surrounding the upper lip                                                                                          |
| <b>levator palpebrae superioris muscle</b> (lih-VAYT-ur PAL-puh-bree soo-peer-ee-OR-is MUS-uhl) | p. 154 | a thin muscle that controls the eyelid                                                                                    |
| <b>lymph</b> (LIMF)                                                                             | p. 165 | colorless, watery fluid that circulates in the lymphatic system; carries waste and impurities from cells                  |
| <b>lymph capillaries</b>                                                                        | p. 165 | blind-end tubes occurring individually or in clusters that are the origin of lymphatic vessels                            |
| <b>lymph nodes</b>                                                                              | p. 165 | glandlike structures found inside lymphatic vessels that filter lymph                                                     |
| <b>lymphatic/immune system</b> (lim-FAT-ik/ih-MYOOON SIS-tum)                                   | p. 165 | consists of lymph, lymph nodes, the thymus gland, the spleen, and lymph vessels that act as an aid to the blood system    |
| <b>mandible</b> (MAN-duh-bul)                                                                   | p. 150 | lower jawbone                                                                                                             |
| <b>mandibular nerve</b> (man-DIB-yuh-lur NURV)                                                  | p. 159 | branch of the fifth cranial nerve that affects the muscles of the chin, lower lip, and external ear                       |
| <b>marginal mandibular nerve</b> (MAR-jin-ul man-DIB-yuh-lur NURV)                              | p. 160 | affects the muscles of the chin and lower lip                                                                             |
| <b>masseter</b> (muh-SEET-ur)                                                                   | p. 153 | one of the jaw muscles used in chewing                                                                                    |
| <b>maxillae</b> (mak-SIL-ee)                                                                    | p. 150 | bones of the upper jaw                                                                                                    |
| <b>maxillary nerve</b> (MAK-suh-lair-ee NURV)                                                   | p. 159 | supplies impulses to the upper part of the face                                                                           |
| <b>mental nerve</b> (MEN-tul NURV)                                                              | p. 159 | nerve that affects the skin of the lower lip and chin                                                                     |
| <b>mentalis muscle</b> (men-TAY-lis MUS-uhl)                                                    | p. 154 | muscle that elevates the lower lip and raises and wrinkles the skin of the chin                                           |
| <b>metacarpals</b> (met-uh-KAR-pulz)                                                            | p. 151 | the bones of the palm, consisting of five slender bones between the carpus and the phalanges                              |
| <b>middle temporal artery</b> (MID-ul TEM-puh-rul ART-uh-ree)                                   | p. 164 | artery that supplies blood to the temples                                                                                 |
| <b>mitosis</b> (my-TOH-sis)                                                                     | p. 146 | cells dividing into two new cells (daughter cells)                                                                        |
| <b>motor nerves</b> (MOH-tur NURVZ)                                                             | p. 158 | nerves that carry impulses from the brain to the muscles                                                                  |

|                                                                             |        |                                                                                                                                                                                     |
|-----------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>muscle tissue</b>                                                        | p. 147 | contracts and moves various parts of the body                                                                                                                                       |
| <b>muscular system</b> (MUS-kyuh-lur SIS-tum)                               | p. 151 | body system that covers, shapes, and supports the skeletal tissue                                                                                                                   |
| <b>myology</b> (my-AHL-uh-jee)                                              | p. 151 | study of the structure, function, and diseases of the muscles                                                                                                                       |
| <b>nasal bones</b> (NAY-zul BONZ)                                           | p. 150 | bones that form the bridge of the nose                                                                                                                                              |
| <b>nasal nerve</b> (NAY-zul NURV)                                           | p. 159 | nerve that affects the point and lower sides of the nose                                                                                                                            |
| <b>nerve tissue</b>                                                         | p. 147 | type of tissue that carries messages to and from the brain and controls and coordinates all bodily functions                                                                        |
| <b>nerves</b>                                                               | p. 158 | whitish cords made up of bundles of nerve fibers held together by connective tissue, through which impulses are transmitted                                                         |
| <b>neuron</b> (NOO-rah-n)                                                   | p. 158 | also known as <i>nerve cell</i> ; the primary structural unit of the nervous system, consisting of the cell body, nucleus, dendrites, and the axon                                  |
| <b>nucleus</b> (NOO-klee-us)                                                | p. 145 | dense, active protoplasm found in the center of a cell; important to reproduction and metabolism                                                                                    |
| <b>occipital artery</b> (ahk-SIP-it-ul AR-tuh-ree)                          | p. 164 | artery that supplies the scalp and back of the head up to the crown                                                                                                                 |
| <b>occipital bone</b> (ahk-SIP-it-ul BOHN)                                  | p. 149 | hindmost bone of the skull; located below the parietal bones                                                                                                                        |
| <b>occipitalis</b> (ahk-SIP-ih-tahl-is)                                     | p. 152 | back of the epicranium; muscle that draws the scalp backward                                                                                                                        |
| <b>ophthalmic nerve</b> (ahf-THAL-mik NURV)                                 | p. 159 | supplies impulses to the skin of the forehead, upper eyelids, and interior portion of the scalp, orbit, eyeball, and nasal passage                                                  |
| <b>orbicularis oculi muscle</b> (or-bik-yuh-LAIR-is AHK-yuh-lye MUS-uhl)    | p. 154 | ring muscle of the eye socket                                                                                                                                                       |
| <b>orbicularis oris muscle</b> (or-bik-yuh-LAIR-is OH-ris MUS-uhl)          | p. 154 | flat band around the upper and lower lips                                                                                                                                           |
| <b>organs</b>                                                               | p. 147 | structures composed of specialized tissues performing specific functions                                                                                                            |
| <b>origin</b> (OR-ih-jin)                                                   | p. 151 | more fixed part of a muscle that does not move                                                                                                                                      |
| <b>os</b> (AHS)                                                             | p. 148 | means “bone”; it is used as a prefix in many medical terms                                                                                                                          |
| <b>osteology</b> (ahs-tee-AHL-oh-jee)                                       | p. 148 | the study of the anatomy, structure, and function of the bones                                                                                                                      |
| <b>parietal artery</b> (puh-RY-ate-ul ART-uh-ree)                           | p. 164 | artery that supplies blood to the side and crown of the head                                                                                                                        |
| <b>parietal bones</b> (puh-RY-ate-ul BONZ)                                  | p. 149 | bones that form the sides and top of the cranium                                                                                                                                    |
| <b>pectoralis major</b> (pek-tor-AL-is MAY-jor) and <b>pectoralis minor</b> | p. 155 | muscles of the chest that assist the swinging movements of the arm                                                                                                                  |
| <b>peripheral nervous system (PNS)</b> (puh-RIF-uh-rul NUR-vus SIS-tum)     | p. 157 | made up of sensory and motor nerve fibers that extend from the brain and spinal cord to all parts of the body. Their function is to carry impulses or messages to and from the CNS. |
| <b>phalanges</b> (FA-lanj-eez)                                              | p. 151 | bone of the fingers or toes                                                                                                                                                         |
| <b>physiology</b> (fiz-ih-OL-oh-jee)                                        | p. 144 | study of the functions or activities performed by the body’s structures                                                                                                             |
| <b>platysma</b> (plah-TIZ-muh)                                              | p. 153 | broad muscle extending from the chest and shoulder muscles to the side of the chin that is responsible for depressing the lower jaw and lip                                         |



|                                                                                  |        |                                                                                                                                                                    |
|----------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>posterior auricular artery</b><br>(poh-STEER-ee-ur aw-RIK-yuh-lur AR-tuh-ree) | p. 164 | artery that supplies blood to the scalp, behind and above the ear                                                                                                  |
| <b>posterior auricular nerve</b><br>(poh-STEER-ee-ur aw-rik-yuh-lur NURV)        | p. 160 | nerve that affects the muscles behind the ear at the base of the skull                                                                                             |
| <b>procerus</b> (proh-SEE-rus)                                                   | p. 154 | muscle that covers the bridge of the nose, depresses the eyebrows, and wrinkles the nose                                                                           |
| <b>pronator</b> (proh-NAY-tohr)                                                  | p. 156 | muscle that turns the hand inward so that the palm faces downward.                                                                                                 |
| <b>protoplasm</b> (PROH-toh-plaz-um)                                             | p. 145 | a colorless jellylike substance found inside cells in which food elements such as proteins, fats, carbohydrates, mineral salts, and water are present              |
| <b>pulmonary circulation</b> (PUL-muh-nayr-ee sur-kyoo-LAY-shun)                 | p. 161 | takes deoxygenated blood to the lungs for oxygenation and waste removal and then returns that blood to the heart so oxygen-rich blood can be delivered to the body |
| <b>radius</b> (RAY-dee-us)                                                       | p. 151 | smaller bone in the forearm on the same side as the thumb                                                                                                          |
| <b>reflex</b> (REE-fleks)                                                        | p. 158 | automatic nerve reaction to a stimulus                                                                                                                             |
| <b>reproductive system</b> (ree-proh-DUK-tiv SIS-tum)                            | p. 167 | body system responsible for processes by which plants and animals produce offspring                                                                                |
| <b>respiratory system</b><br>(RES-puh-rah-tor-ee SIS-tum)                        | p. 148 | consists of the lungs and air passages; enables breathing                                                                                                          |
| <b>risorius muscle</b> (rih-ZOR-ee-us MUS-uhl)                                   | p. 154 | muscle of the mouth that draws the corner of the mouth out and back as in grinning                                                                                 |
| <b>sensory nerves</b>                                                            | p. 158 | nerves that carry impulses or messages from the sense organs to the brain                                                                                          |
| <b>serratus anterior</b> (ser-RAT-us an-TEER-ee-ur)                              | p. 155 | muscle of the chest that assists in breathing and in raising the arm                                                                                               |
| <b>seventh cranial nerve</b> (SEV-AHNTH CRAN-ee-ahl NURV)                        | p. 159 | chief motor nerve of the face                                                                                                                                      |
| <b>skeletal system</b> (SKEL-uh-tul SIS-tum)                                     | p. 148 | physical foundation of the body; composed of bones and movable and immovable joints                                                                                |
| <b>smaller occipital nerve</b><br>(SMAWL-ur ahk-SIP-ut-ul NURV)                  | p. 160 | nerve that affects the scalp and muscles behind the ear                                                                                                            |
| <b>sphenoid bone</b> (SFEE-noyd BOHN)                                            | p. 149 | bone that connects all the bones of the cranium                                                                                                                    |
| <b>spinal cord</b> (SPY-nal KORD)                                                | p. 157 | portion of the central nervous system that originates in the brain and runs downward through the spinal column                                                     |
| <b>sternocleidomastoideus</b> (STUR-noh-KLEE-ih-doh-mas-TOYD-ee-us)              | p. 153 | muscle of the neck that depresses and rotates the head                                                                                                             |
| <b>submental artery</b> (sub-MEN-tul AR-tuh-ree)                                 | p. 164 | artery that supplies blood to the chin and lower lip                                                                                                               |
| <b>superficial temporal artery</b><br>(soo-pur-FISH-ul TEM-puh-rul AR-tuh-ree)   | p. 164 | artery that supplies blood to the muscles of the front, sides, and top of the head                                                                                 |

|                                                                       |        |                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>superior labial artery</b> (soo-PEER-ee-ur LAY-bee-ul AR-tuh-ree)  | p. 164 | artery that supplies blood to the upper lip and lower region of the nose                                                                                                                                                                                                                               |
| <b>supinator</b> (SOO-puh-nayt-ur)                                    | p. 156 | muscle of the forearm that rotates the radius outward and the palm upward                                                                                                                                                                                                                              |
| <b>supraorbital artery</b> (soo-pruh-OR-bih-tul AR-tuh-ree)           | p. 164 | artery that supplies blood to the upper eyelid and forehead                                                                                                                                                                                                                                            |
| <b>supraorbital nerve</b> (soo-pruh-OR-bih-tul NURV)                  | p. 159 | nerve that affects the skin of the forehead, scalp, eyebrows, and upper eyelids                                                                                                                                                                                                                        |
| <b>supratrochlear nerve</b> (soo-pruh-TRAHK-lee-ur NURV)              | p. 159 | nerve that affects the skin between the eyes and the upper side of the nose                                                                                                                                                                                                                            |
| <b>systemic circulation</b> (sis-TEM-ik sur-kyoo-LAY-shun)            | p. 161 | carries the oxygen-rich blood from the heart throughout the body and returns deoxygenated blood back to the heart                                                                                                                                                                                      |
| <b>temporal bones</b> (TEM-puh-rul BONZ)                              | p. 149 | bones that form the sides of the head in the ear region                                                                                                                                                                                                                                                |
| <b>temporal nerve</b> (TEM-poh-rul NURV)                              | p. 160 | nerve that affects the muscles of the temple, side of the forehead, eyelid, eyebrow, and upper cheek                                                                                                                                                                                                   |
| <b>temporalis</b> (tem-poh-RAY-lis)                                   | p. 153 | muscle that aids in opening and closing the mouth and chewing                                                                                                                                                                                                                                          |
| <b>thorax</b> (THOR-aks)                                              | p. 150 | the chest                                                                                                                                                                                                                                                                                              |
| <b>tissue</b> (TISH-oo)                                               | p. 147 | collections of similar cells that perform a particular function                                                                                                                                                                                                                                        |
| <b>transverse facial artery</b> (tranz-VURS FAY-shul AR-tuh-ree)      | p. 164 | artery that supplies blood to the skin and the masseter                                                                                                                                                                                                                                                |
| <b>trapezius</b> (truh-PEE-zee-us)                                    | p. 155 | muscle that covers the back of the neck and upper and middle region of the back                                                                                                                                                                                                                        |
| <b>triangularis muscle</b> (try-ang-gyuh-LAY-rus MUS-uhl)             | p. 154 | muscle that extends alongside the chin and pulls down the corner of the mouth                                                                                                                                                                                                                          |
| <b>tricep</b> (TRY-sep)                                               | p. 156 | large muscle that covers the entire back of the upper arm and extends the forearm                                                                                                                                                                                                                      |
| <b>ulna</b> (UL-nuh)                                                  | p. 151 | inner and larger bone of the forearm                                                                                                                                                                                                                                                                   |
| <b>veins</b> (VAYNS)                                                  | p. 161 | thin-walled blood vessels that are less elastic than arteries; veins contain cuplike valves that keep blood flowing in one direction to the heart and prevent blood from flowing backward. Veins carry blood containing waste products back to the heart and lungs for cleaning and to pick up oxygen. |
| <b>venules</b> (VEEN-yools)                                           | p. 161 | small vessels that connect the capillaries to the veins; they collect blood from the capillaries and drain it into the veins                                                                                                                                                                           |
| <b>zygomatic bones</b> (zy-goh-MAT-ik BONZ)                           | p. 150 | bones that form the prominence of the cheeks                                                                                                                                                                                                                                                           |
| <b>zygomatic nerve</b> (zy-goh-MAT-ik NURV)                           | p. 160 | affects the muscle of the upper part of the cheek                                                                                                                                                                                                                                                      |
| <b>zygomaticus major muscles</b> (zy-goh-MAT-ih-kus MAY-jor MUS-uhlz) | p. 155 | muscle extending from the zygomatic bone to the angle of the mouth; pull the mouth upward and backward as in laughing                                                                                                                                                                                  |
| <b>zygomaticus minor muscles</b> (zy-goh-mat-ih-kus MY-nor MUS-uhlz)  | p. 155 | muscles on both sides of the face that extend from the zygomatic bone to the upper lips; pull the upper lip backward, upward, and outward, as when you are smiling                                                                                                                                     |

7



BASICS OF CHEMISTRY



## CHAPTER OUTLINE

### INTRODUCTION

#### WHY STUDY BASICS OF CHEMISTRY?

#### UNDERSTAND BASIC CHEMISTRY

#### DISCUSS THE PROPERTIES OF WATER AND pH

#### IDENTIFY COSMETIC PREPARATIONS USED IN BARBERING

# LEARNING OBJECTIVES

After completing this chapter, you will be able to:

### LO<sup>1</sup>

Define *organic* and *inorganic chemistry*.

### LO<sup>2</sup>

Define the properties of *matter*.

### LO<sup>3</sup>

Discuss the physical and chemical properties of matter.

### LO<sup>4</sup>

Explain oxidation-reduction reactions.

### LO<sup>5</sup>

Describe emulsions, suspensions, and solutions.

### LO<sup>6</sup>

Define *pH* and describe the pH scale.

### LO<sup>7</sup>

Explain how product pH levels affect the hair and skin.

### LO<sup>8</sup>

Name nine types of shampoos.

### LO<sup>9</sup>

List four classifications of conditioners.

### LO<sup>10</sup>

Recognize other cosmetic preparations used in barbering services.

# Introduction

This chapter provides an overview of basic chemistry so that you will understand its applications to the field of barbering. Barbers use chemical products on a daily basis and it is important that you know how to use these preparations safely and effectively in the performance of your work.

One of the most important uses of chemicals in the barbershop is the application of cleaners and disinfectants to maintain an effective infection control program. Products are usually more economical if purchased in concentrated form and will require you to mix the solutions; others will be used straight from the container. Always follow the manufacturer's directions for mixing and/or use. Hair care and skin care products are also chemical preparations that are used daily in the barbershop. Since the ingredients in these products can vary greatly, it is important that you understand their properties so that you can make informed selections for your clients' services.

Chemical preparations are necessary when a permanent change to the hair structure is desired. Certain haircolor formulations, permanent waving lotions, and chemical hair-relaxing products all create permanent changes within the hair; in fact, without chemicals, a permanent change in the hair is not possible!

why study

## BASICS OF CHEMISTRY?

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**Barbers should study and have a thorough understanding of the basics of chemistry as it applies to barbering because:**

- An understanding of chemistry will enable you to use professional products effectively and safely.
- Chemicals are common ingredients found in the products used in the barbershop and in barbering services.
- An understanding of chemical preparations and their effect on the hair and skin will help you select the best products to achieve the desired outcome.
- An understanding of chemical preparations and their effect on the hair and skin will help you prevent or solve problems that can occur during product applications.

## Understand Basic Chemistry

After reading this section, you will be able to:

- LO 1 Define *organic* and *inorganic chemistry*.
- LO 2 Define the properties of *matter*.
- LO 3 Discuss the physical and chemical properties of matter.
- LO 4 Explain oxidation-reduction reactions.
- LO 5 Describe emulsions, suspensions, and solutions.



## EXPLAIN THE DIFFERENCES BETWEEN ORGANIC AND INORGANIC CHEMISTRY

**Chemistry** (KEM-is-tree) is the science that deals with the composition, structure, and properties of matter and how matter changes under different chemical conditions. The field is divided into two areas: organic and inorganic chemistry.

**Organic chemistry** (or-GAN-ik KEM-is-tree) is the study of substances that contain the element carbon; all living things are made of compounds that contain carbon. The term *organic* applies to all living things and those things that were once alive. Gasoline, synthetic fabrics, plastics, and pesticides are all considered organic because they are manufactured from natural gas and oil, which are the remains of plants and animals that died millions of years ago. Most organic substances will burn.

**Inorganic chemistry** (in-or-GAN-ik KEM-is-tree) is the study of substances that do not contain carbon but may contain hydrogen. Inorganic substances are not, and never were, alive; therefore, they will not burn. Metals, minerals, water, air, and ammonia are inorganic substances.

## DEFINE MATTER

**Matter** (MAT-ur) is defined as anything that occupies space (volume) and has mass (weight). All matter has physical and chemical properties and exists in the form of a solid, liquid, or gas.

Matter has *physical* properties we can touch, taste, smell, or see, but not everything is matter. For example, we can see visible light and electrical sparks, but light and electricity are forms of energy and energy is not matter because it does not have mass; in fact, energy can be defined as the ability to change or put matter into motion. The *chemical* properties of matter can only be seen or measured when a chemical change or chemical reaction occurs.

## Elements

An **element** (EL-uh-ment) is the simplest form of chemical matter and contains only one type of atom. It cannot be broken down into a simpler substance without a loss of identity. Refer to **Figure 7-1** as you read through the following:

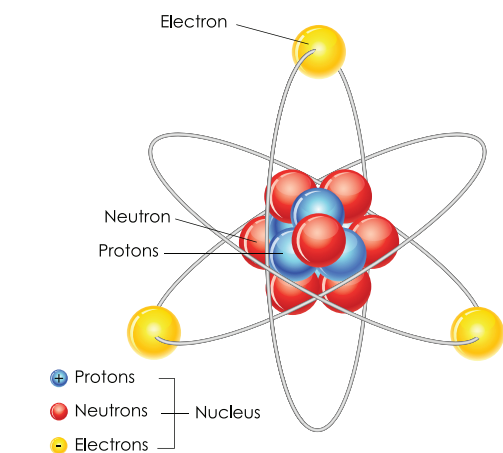
- There are 118 different elements known to science today with 98 of these occurring naturally on Earth. The remaining elements, known as *synthetic elements*, are produced artificially or through synthesis.

**figure 7-1**

Periodic table information for carbon.

|               |        |             |  |
|---------------|--------|-------------|--|
| Atomic Number | 6      |             |  |
|               | C      | Symbol      |  |
| Element Name  | Carbon |             |  |
|               | 12.01  | Atomic Mass |  |

figure 7-2  
Atom.



## Atom Structure

- All the matter in the known universe is made up of elements that have their own distinct physical and chemical properties. Each element is given a letter symbol such as O for oxygen, S for sulfur, or H for hydrogen.
- Elements are made up of atoms. The atomic number of an element indicates how many protons are in one atom of that element. If the number of protons changes, the element changes.
- The complete Periodic Table of Elements can be found in chemistry textbooks or by searching the Internet.

### Atoms

**Atoms** (AT-umz) are the basic building blocks of all matter and are the smallest particle of an element that has the chemical identity of the chemical. At the center of an atom is the nucleus that is made up of subatomic particles of protons (P), with a positive charge, and neutrons (N), with no charge. The nucleus is orbited by electrons (E) with a negative charge (Figure 7-2). As the basic unit of matter, atoms cannot be divided into simpler substances by ordinary chemical means.

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### Molecules

A **molecule** is formed when two or more atoms combine chemically in definite (fixed) proportions. For example, hydrogen ( $H_2$ ) contains two atoms of hydrogen. The air, or atmospheric gases, we breathe is primarily made up of nitrogen, oxygen, carbon dioxide, argon, and water vapor. The oxygen in air is an example of an elemental molecule. **Elemental molecules** (EL-uh-ment-uhl MAHL-uh-kyoolz) consist of two or more atoms of the same element in fixed proportions; in this case, two atoms of the element oxygen, which is written as  $O_2$ . Ozone ( $O_3$ ), a major component of smog, is another elemental molecule but contains three atoms of oxygen (Figure 7-3).

**Compound molecules** (KAHM-pownd MAHL-uh-kyoolz) are chemical combinations of two or more atoms of different elements. Sodium chloride, or common table salt, is a compound molecule that is a chemical combination of one atom of sodium (Na) and one atom of chlorine (Cl). Other examples of compound molecules you will recognize are water ( $H_2O$ ), carbon dioxide ( $CO_2$ ), and hydrogen peroxide ( $H_2O_2$ ) (Figure 7-4).

figure 7-3  
Elemental molecules.

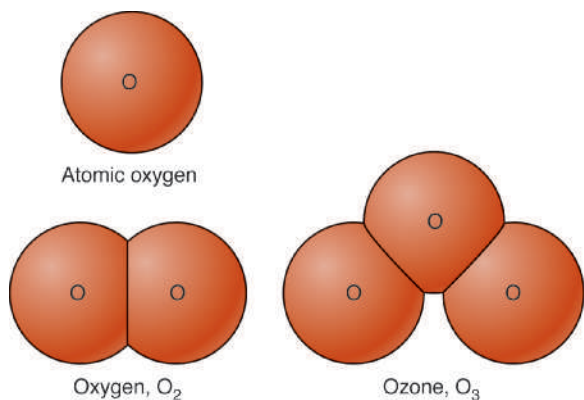


figure 7-4  
Compound molecules.

